Does Modelling Driver Adaptation Change Charging Infrastructure Decisions? A Comparison of Data Replay and Agent-Based Simulation for Minibus-Taxi Electrification in Cape Town

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Abstract

Electrifying minibus taxis requires charging infrastructure that accommodates both vehicle energy requirements and the operational decisions of drivers. An existing planning approach uses data-replay models to assess whether observed mobility is compatible with electrification. However, it remains unclear whether introducing explicit representations of driver agency changes the charging infrastructure that is selected. This study compares a data-replay model with a decision-adaptive agent-based model for three minibus-taxi communities in Cape Town, South Africa, selected from eight communities identified using vehicle GPS telemetry. Both models electrify the same observed vehicles, while the agent-based model also represents synthetic internal combustion engine vehicles competing for passengers on the same corridors. Every candidate charging design is evaluated using the model-specific measures of operational performance.

Under a common reference design selected by the data-replay model, 57.0%, 22.6% and 49.0% of vehicle-days end in stranding in Khayelitsha, Bellville and Wynberg, respectively. No validated agent-based run strands, because drivers can travel to charging stations. However, under the reference design, the electrified agents carry 18.5%, 6.0% and 23.8% fewer passengers than in their internal-combustion counterfactuals. When each model selects its own design, the agent-based model increases installed charging power from 250 to 1,000 kW in Khayelitsha, retains 200 kW but reallocates the connectors in Bellville, and increases installed power from 100 to 150 kW in Wynberg. When evaluated in the agent-based model, the data-replay designs produce lower EV passenger throughput than the optimised agent-based designs in all three communities. These differences persist across the tested sensitivity cases.

The results show that replay-based feasibility assessments do not necessarily identify the charging design that best supports passenger service when drivers can adapt. Agent-based modelling captures this additional operational response, but requires substantially more computation and explicit assumptions about driver behaviour and passenger demand. The appropriate modelling approach therefore depends on the planning question and the operational constraints of the proposed electrification pilot.

Keywords: paratransit, minibus taxi, electrification, agent-based modelling, data replay, charging infrastructure, vehicle communities, mixed fleet, multi-objective design evaluation, Cape Town

1. Introduction

TODO: Complete introduction.

1.1. Paratransit electrification as a coupled planning problem

Paratransit systems, such as minibus taxis, are a critical component of urban mobility in many cities, particularly in developing countries. As global efforts to reduce greenhouse gas emissions intensify, the electrification of these systems has become a focal point for sustainable urban transport planning [1, 2].

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However, the transition from internal combustion engine (ICE) vehicles to electric vehicles (EVs) introduces a complex set of interdependencies that were previously weakly coupled. Specifically, the mobility of vehicles is now closely linked to vehicle drive-train efficiency, battery capacity, and charging infrastructure, which in turn affects the electrical grid and drivers' ability to service passenger demand [3]. This coupling creates new challenges for planning and operation, as the limitations of electric vehicles (EVs), such as slower charging times and limited range, changes the operational status quo, introducing opportunity costs that were not present for internal combustion engine (ICE) vehicles []. When investigating the electrification of paratransit system, it is therefore essential that the planning models account for the interactions between these systems, as well as the adaptive behaviour of drivers who respond to these constraints in their operational decisions [].

In South Africa, minibus-taxi drivers have significant agency over their operational decisions, including where, when, and how long to operate []. This agency allows drivers to adapt their behaviour in response to the constraints imposed by electrification, such as limited range and charging availability. As a result, the success of electrification initiatives is not solely dependent on the technical feasibility of EVs but also on how drivers respond to these new operational constraints. Understanding and modelling driver behaviour is therefore crucial for accurately predicting the outcomes of electrification and for designing effective charging infrastructure that meets both operational and passenger demand requirements.

Therefore, the planning of charging infrastructure for paratransit electrification must consider whether mobility should be treated as fixed or adaptive. Fixed-mobility models (M_{R}), otherwise termed "data-replay models", assume that observed vehicle activity will largely persist after electrification, while adaptive-mobility models (M_{A}) account for the potential changes in driver behaviour in response to introduced constraints. Where the former may provide a more straightforward assessment of electrification feasibility, the latter accounts for the endogenous changes in driver behaviour, offering a more nuanced understanding of how drivers may alter their operations, which can significantly impact the effectiveness of charging infrastructure to support the servicing passenger demand.

1.2. Modelling paratransit: the tale of two paradigms

Modelling the electrification of paratransit systems is a complex problem that requires careful consideration of both technical and behavioural factors. In the domain of assessing the integration of electric vehicles into paratransit systems, two primary modelling paradigms have emerged: fixed-mobility data-replay models and adaptive-mobility agent-based models. Each paradigm offers distinct advantages and limitations, and the choice between them can significantly impact the predicted outcomes of electrification initiatives.

Planning models are normative tools that shape transitions and encode assumptions about which outcomes planners can observe and which they cannot [].

Because of significant economical, infrastructural, and operational barriers, realistically, the first electric minibus taxis will be a small group of vehicles that share stops, routes, and hubs, rather than an entire electrified fleet. This means that, for the purposes of this study, the planning and evaluation of electrification initiatives are conducted at the community scale, where the interactions between electric and internal combustion engine (ICE) vehicles are particularly relevant. In such a mixed-fleet environment, the time spent by electric vehicles at charging stations represents an opportunity cost, as ICE competitors can continue to carry passengers during these periods. Furthermore, vehicle communities are not necessarily aligned with taxi associations or licensed routes, and the study does not assign any association to a vehicle or community. Therefore, the comparison of fixed-mobility data-replay models and adaptive-mobility agent-based models is most relevant at the scale and in the market in which electrification pilots are likely to occur.

1.3. Research gap

This research investigates the implications of driver agency for the planning of paratransit electrification pilots. Existing techniques can evaluate whether observed paratransit mobility is compatible with electrification [], while behaviourally explicit models can represent adaptive responses to energy and charging constraints []. What remains unclear is whether these different representations lead to materially different infrastructure planning decisions when applied to the same empirically observed vehicles, and whether infrastructure selected under fixed-mobility assumptions remains effective once drivers respond endogenously and compete for passengers with the internal-combustion vehicles that share their corridors. It is also unclear whether such differences are consistent across the distinct operating communities in which electrification pilots are likely to take place.

1.4. Research questions

This study poses and addresses the following research questions:

1. For the same observed community vehicles, how do predicted electrification outcomes differ when mobility is represented as fixed empirical activity (M_{R}) versus endogenous driver behaviour (M_{A}) under a common reference charging design?
2. Does endogenous driver behaviour change the charging infrastructure that is selected when each model is evaluated using the outcomes it can validly represent?
3. What does a replay-derived charging design cost in passenger service when it is evaluated in the agent-based model, and what share of passengers do the electric vehicles lose to internal-combustion competitors while they charge?
4. Do these answers point in the same direction in three different vehicle communities, and how sensitive are they to the corridor population and the vehicle assumptions?

1.5. Contributions

This study examines the implications of explicit behavioural representations of driver agency for paratransit electrification planning. It contributes to the literature in five ways. First, it performs a controlled comparison of fixed-mobility data replay and decision-adaptive agent-based modelling, in which the same observed vehicles are electrified in both models, in three minibus-taxi communities in Cape Town. Second, it introduces the vehicle community, found from router GPS data and screened for electrification by a rule that is frozen before any agent-based run, as a reproducible unit for the selection of electrification pilots. Third, it simulates the full corridor population in the agent-based model, with electric vehicles in a mixed fleet, which makes the passengers that electric vehicles lose to internal-combustion competitors measurable. Fourth, it evaluates every charging design in both models, so the Pareto fronts are exact and the comparison does not depend on the performance of a search algorithm. Fifth, the comparison framework adheres to model-native objective spaces, rather than forcing a single objective space onto models with different behavioural assumptions.

2. Related work

TODO: Complete literature review.

2.1. Electrification feasibility under observed mobility

2.2. Behavioural modelling of paratransit operations

2.3. Charging infrastructure planning under operational uncertainty

2.4. Reconstructing paratransit networks and operating communities from GPS data

3. Study area and data

3.1. Minibus-taxi operation in Cape Town

Minibus taxis carry the largest share of the public-transport trips in Cape Town []. Their operation is shaped by vehicle owners, route associations and the provincial operating-licence system []. Five operating characteristics are particularly relevant to electrification and to the models developed in this study.

First, passengers commonly board at taxi ranks, where vehicles queue on a first-in, first-out basis and depart when sufficiently full []. This is known as fill-and-go loading. Second, passengers may board or alight at stops along a route, although boarding is generally concentrated at terminals. Third, vehicles tend to operate on a limited set of routes between established terminals []. Fourth, drivers may wait or move slowly along a route to find passengers, a practice referred to here as hold-back []. Finally, demand is directional and varies over the day []. During the morning peak, many passengers travel from residential areas towards employment centres, leaving some taxis with relatively few passengers on the return journey []. The pattern often reverses in the evening.

These features create specific charging constraints. Long dwell periods at ranks and terminals provide potential charging opportunities, but a taxi that leaves a loading queue to charge may lose its position and

the passengers it would otherwise have carried. The models therefore need to represent both the availability of charging opportunities and the cost of using them.

Shown in Figure 1a, the study area is the union of the City of Cape Town metropolitan municipality and the surrounding Stellenbosch and Drakenstein municipalities. Telemetry data outside the area is flagged but not deleted. Around 19% of the driven distance lies outside he study area, and a vehicle-day with 50 km or more outside the area is considered a long-distance day.

3.2. Vehicle telemetry

The observed fleet dataset is a subset of the Sebenza router telemetry [4], which is a GPS dataset of minibus taxis in South Africa. The subset consisted of 378 minibus taxis operating in Cape Town over 11 weekdays from 1 to 15 December 2025, resulting in a total of 3,916 vehicle-days. The second half of December was excluded because of the school holidays and the Christmas period which produce a distinct mobility regime [5]. Each GPS point includes a pseudonymised vehicle identifier, a timestamp, a position, and a quality flag. The median sampling interval is 79 seconds, with the 5th to 95th percentile ranging from 60 to 81 seconds. On average, a moving taxi covers a median distance of 435 meters between two points. While the start, end, and overnight positions of each day are observed, stops shorter than 2 to 3 minutes, which include most boarding and alighting events along a route, are not captured in the dataset.

Table 1 summarises the duty cycle of the observed Cape Town fleet. The median valid vehicle-day covers 190 km (interquartile range 130 to 247 km), whereas the battery window of the study EV gives about 130 km; only 24.9% of the valid days fall within this range. The median driving window, from the departure after the first stop of the day to the arrival at the last, is 17.4 h. The median taxi therefore parks for about 6.6 h, in which a 3.7 kW socket adds at most about 22 kWh, less than a third of the median daily need of 72 kWh. Further analysis shows that long dwells concentrate at ranks, official roue termini, and stations. Additionally, each vehicle varies strongly from day to day: the intraclass correlation is 0.43 (95% confidence interval 0.36 to 0.50) for the daily distance and 0.62 (0.56 to 0.67) for the daily moving time, thus 57% of the variance of the daily distance lies between the days of one vehicle.

Table 1: Duty cycle of the Cape Town fleet by day class. Long-distance: the share of all vehicle-days with 50 km or more outside the study area. Distance and driving window: over the valid vehicle-days, with the interquartile range in brackets; the driving window runs from the end of the stop that contains the first point of the day to the start of the stop that contains the last point. Within range: the share of the valid vehicle-days of 130 km or less, the distance of the EV battery window.

	Dates		Vehicle-	Valid	Long-dist.	Distance (km)		Driving window (h)	Within
Day class	(Dec. 2025)	Vehicles	days	(%)	(%)	Mean	Median [IQR]	Median [IQR]	range (%)
Term weekdays	1–5, 8–10	374	2,856	61.8	2.5	193	189 [128–242]	17.5 [15.2–19.9]	25.8
School-holiday weekdays	11, 12, 15	367	1,060	64.2	7.5	214	196 [136–259]	17.3 [15.2–19.5]	22.6
All study weekdays	1–15	378	3,916	62.5	3.9	199	190 [130–247]	17.4 [15.2–19.8]	24.9

To maintain privacy, the study uses only pseudonymised data. This paper reports aggregates only and never assigns a taxi association to a vehicle or a community. Additionally, home locations are neither named nor shown on maps.

3.3. Infrastructure, planning and survey data

Table 2 lists the public data that complement the telemetry. The formal stops combine the City of Cape Town route termini, and the taxi ranks and railway stations in OpenStreetMap. The 2023–2028 Comprehensive Integrated Transport Plan (CITP) of the City of Cape Town [6] provides the daily boardings at minibus-taxi facilities, which set the level of passenger demand, and the morning-peak vehicle trips by direction on surveyed corridors, which set its direction. The provincial route register [] gives only an upper bound of the fleet size at a hub, because one licence can authorise many routes.

3.4. Observation period, data quality and analysis sample

Established by Pretorius et al. [8], an operational vehicle-day runs from 04:00 to 03:59 the following morning. This boundary groups overnight charging with the operating cycle that precedes it, since most taxis are parked during this period. A GPS step is treated as evidence of data loss, and the vehicle-day

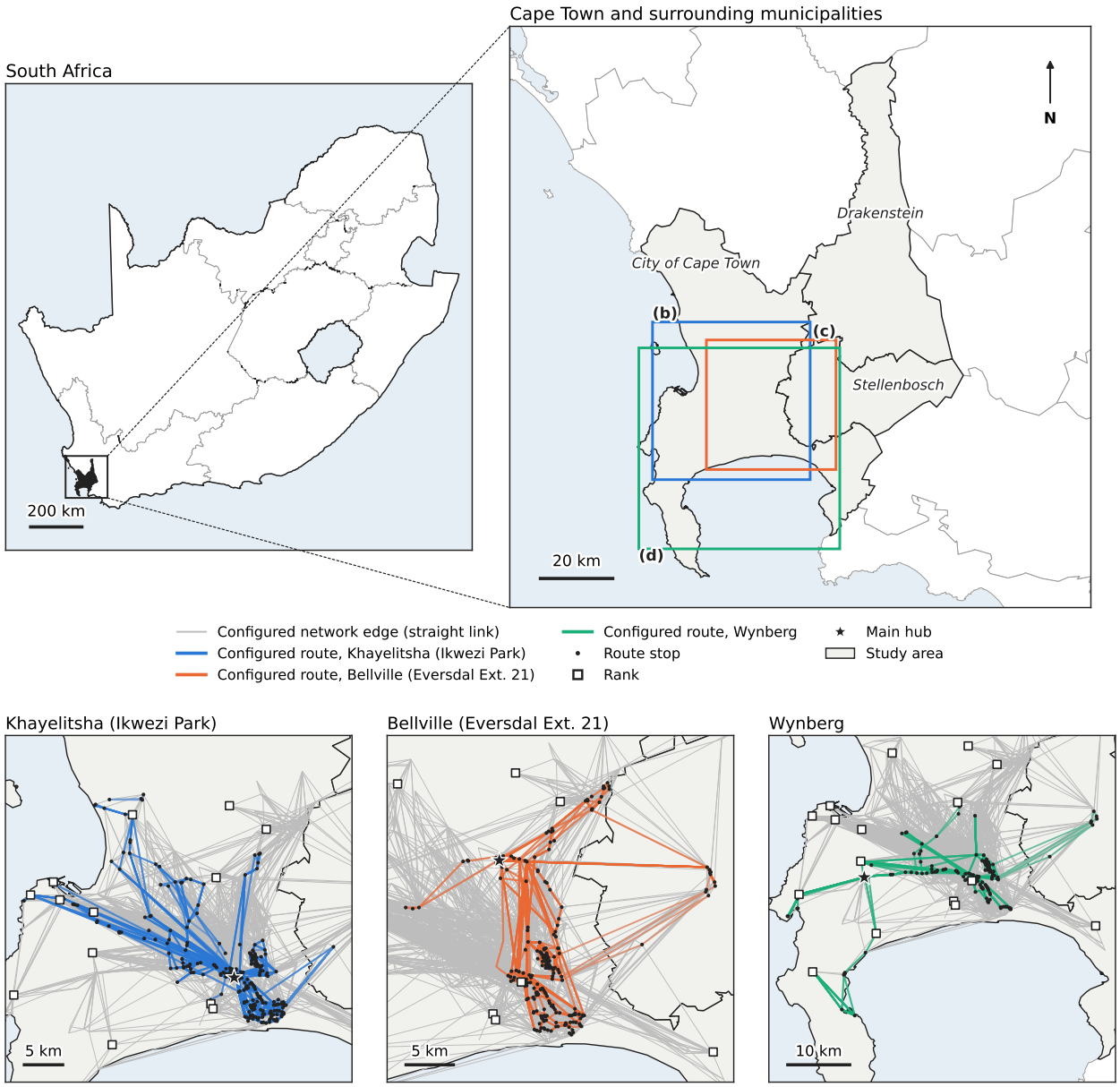


Figure 1: Study context. (a) South Africa with the study area and a zoom on Cape Town and the surrounding municipalities (Census 2011 boundaries [**?**]); the coloured boxes give the extent of (b) to (d). (b) Khayelitsha, (c) Bellville and (d) Wynberg: the configured network of each community as straight links between nodes (grey), the configured routes of the community drivers as straight links between their stops (colour), the route stops (dots), the formal ranks (squares) and the main hub (star). Home nodes and the links to them are not shown.

Table 2: Public data used with the telemetry.

Source	**Content**	**Use**
City of Cape Town (2019) []	Minibus-taxi route termini	Formal nodes
OpenStreetMap, July 2026 []	Taxi ranks and railway stations; drivable roads; land use	Formal nodes; edge lengths; residential ends of corridors
Census 2011 []	Municipalities, main places and subplaces	Study area; place names
CITP, Table 3-9 [6]	Daily boardings at minibus-taxi facilities	Scale factor k
CITP, Table 3-18 [6]	Morning-peak vehicle trips and passengers by direction on surveyed corridors	Load factors; demand audit
City route register (2023) [7]	Valid operating licences by route	Upper-bound check of k
School calendar 2025 []	Last school day (10 December)	Day classes

is marked invalid, when consecutive observations are more than 600 seconds apart and more than 1,000 m apart. A long interval without corresponding movement is instead treated as a parked period.

Of the 3,916 observed vehicle-days, 2,446 (62.5%) satisfy the validity criteria. The valid share varies from 29% to 87% across dates, while the number of active vehicles remains comparatively stable at 351–360 per day. This pattern suggests that variation in the valid share reflects data completeness more than changes in fleet activity. The agent-based model therefore applies a day-level correction factor when scaling observed activity to daily operation.

The study days fall into two classes (Table 1). The eight term weekdays (1 to 5 and 8 to 10 December) end with the last school day, and the three school-holiday weekdays (11, 12 and 15 December) follow it [**?**]. The holiday weekdays have three times the share of long-distance days (7.5% against 2.5%) and a longer mean distance (214 against 193 km), whereas without the long-distance days the two classes are almost the same (mean 184 against 185 km). They therefore represent a different demand regime and are excluded from every fit. Of the term weekdays, Tuesday 2 and Wednesday 10 December, with valid shares of 79% and 86%, are the hold-out days, and the other six are the calibration days. Section 4.1 gives the steps that use each class. No candidate community is excluded on data quality. Their valid shares are 59% to 75%, their gap times 1% to 2%, and their shares of driving outside the network 14% to 47%.

4. Methodology

4.1. Comparative design and shared assumptions

The study evaluates the same observed vehicles using two models, summarised in Table 3. In the data-replay model (M_{R}), each vehicle follows its observed movements. In the agent-based model (M_{A}), drivers generate realistic movements through operational decisions and can respond to energy and charging constraints. The electrified agents in M_{A} compete for passengers with synthetic ICE vehicles operating on the same corridors.

The representation of vehicle movement is the primary difference under investigation. Both models use the same vehicle and charging assumptions, candidate sites and design space, and both electrify the same observed vehicles within each community. Where their operating periods differ, the evaluation windows are made as comparable as the model structures allow. One asymmetry remains: M_{R} allows vehicles to charge during dwell periods that include waiting in a rank loading queue, whereas M_{A} requires drivers to leave the queue to charge. This assumption favours M_{R} and should be considered when interpreting the results.

Table 3: Elements of the two models. Movement is the treatment of the comparison; the other elements are the same in both models, made equivalent, or present in one model only.

Element	M_{R}	M_{A}	**Relation**
Electrified vehicles	The agents of the community	The same agents	Same
Other vehicles	None	Synthetic ICE copies of the agents	M_{A} only
Movement	Observed and fixed	Output of driver decisions	Treatment
Days	The eight observed term weekdays	One typical term weekday, simulated for 14 evaluation days	Equivalent
Vehicle and charging values	One assumption set	The same assumption set	Same
Candidate sites and designs	The first three hubs; (N_1, N_2, N_3, P)	The same	Same
Overnight charging	3.7 kW where the vehicle parks	3.7 kW at the home node	Equivalent
Charging in a rank loading queue	Yes	No	Asymmetry
Passenger service	Not represented	Requests, waits and the passengers of each driver	M_{A} only
Selection rule	Service near the best, then the lowest peak site load	The same principle, with a passenger measure of service	Same principle

4.1.1. Vehicle and charging values

Both models use one assumption set based on the Higer H5C, a 15-seat electric minibus [9]. The vehicle has a usable battery capacity (C) of 70.47 kWh, an empirically measured energy consumption rate of 0.38 kWh/km

in loaded operation [10], a vehicle charging limit of 60 kW and a charging efficiency of 90%. The battery operates between a reserve of $0.2\,C$ and a charge limit of $0.9\,C$. This window holds 49.3 kWh, or about 130 km. Every vehicle charges overnight at 3.7 kW from a socket outlet, and site chargers serve vehicles in a first-come, first-served queue.

4.1.2. Unit of analysis and study days

The unit of analysis is the vehicle community: a group of vehicles with overlapping operating areas, routes and hubs. These shared operating patterns create overlapping charging requirements and make communities a useful scale for evaluating electrification pilots. A community is a data-derived operating group, not a taxi association, fleet operator or licensed route.

Network construction, route inference and community detection use all 11 observation weekdays. Agent-based model calibration uses the six designated calibration days, while the two hold-out days remain unused until the calibrated model has been accepted. Since M_{R} has no fitted behavioural parameters, it replays all eight term weekdays.

4.1.3. Planning principle

The study selects one charging design per model and community. The selection principle is to achieve service performance close to the best observed within the model's feasible design set while minimising the required infrastructure. Installed power, calculated as the number of connectors multiplied by their rated power, serves as a proxy for infrastructure cost.

The service measure depends on the model. In M_{R}, the selection rule uses the share of vehicle-days that remain viable under repeated operation, as described by Pretorius et al. [8]. In M_{A}, it uses the passenger throughput of the electrified vehicles, alongside constraints on expired passenger requests. The precise selection rules are given in Section 4.7.2. Since the models represent different aspects of operation, the selected designs are subsequently evaluated in a common agent-based framework.

4.1.4. Pipeline

Figure 2 summarises the study's methodology. GPS telemetry is used to infer the operational network, construct a route library and identify vehicle communities. A predefined screening rule selects three communities for further analysis (Section 4.2.3). For each community, the same observed vehicles are electrified in both models.

The data-replay model evaluates every candidate charging design and selects a reference design, x_R. The agent-based model searches the same design space using its own passenger-service and infrastructure measures, producing a selected design x_A. Both x_R and x_A are then evaluated in the agent-based model. This cross-evaluation measures the operational consequences of selecting infrastructure under fixed-mobility assumptions. The analysis also examines whether these consequences persist across the three communities and under alternative vehicle energy-consumption, charging-policy and fleet-scale assumptions.

4.2. Operational network, routes and vehicle communities

4.2.1. Stop events and the operational network

Stop events are extracted from the GPS trajectories with an anchored-dwell algorithm that tolerates isolated positioning errors. An event has at least three inlier points within 30 m of its anchor and lasts at least 180 s. At the sampling interval of about 80 s, stops shorter than 2 to 3 min, which include most boarding and alighting along a route, are not observable. The home of a vehicle is the location within 300 m of which the first and last points of at least 75% of its valid vehicle-days lie.

One operational network is inferred and applied to all communities. Its nodes come from evidence of stopping where grid cells with stop events from at least five vehicle-days and three vehicles, merged by complete linkage within 200 m, together with the formal ranks, termini and stations that the fleet uses. Its directed edges come from evidence of driving which constitute hops between nodes on at least two vehicle-days, their reverse edges, and links between nodes within 300 m. Edge lengths are shortest drivable paths on OpenStreetMap. After one manual review of 211 flagged items, the network has 726 nodes, of which 18 are ranks. It captures 76% of the stop events and represents 66.5% of the distance driven between stops with respect to the full fleet.

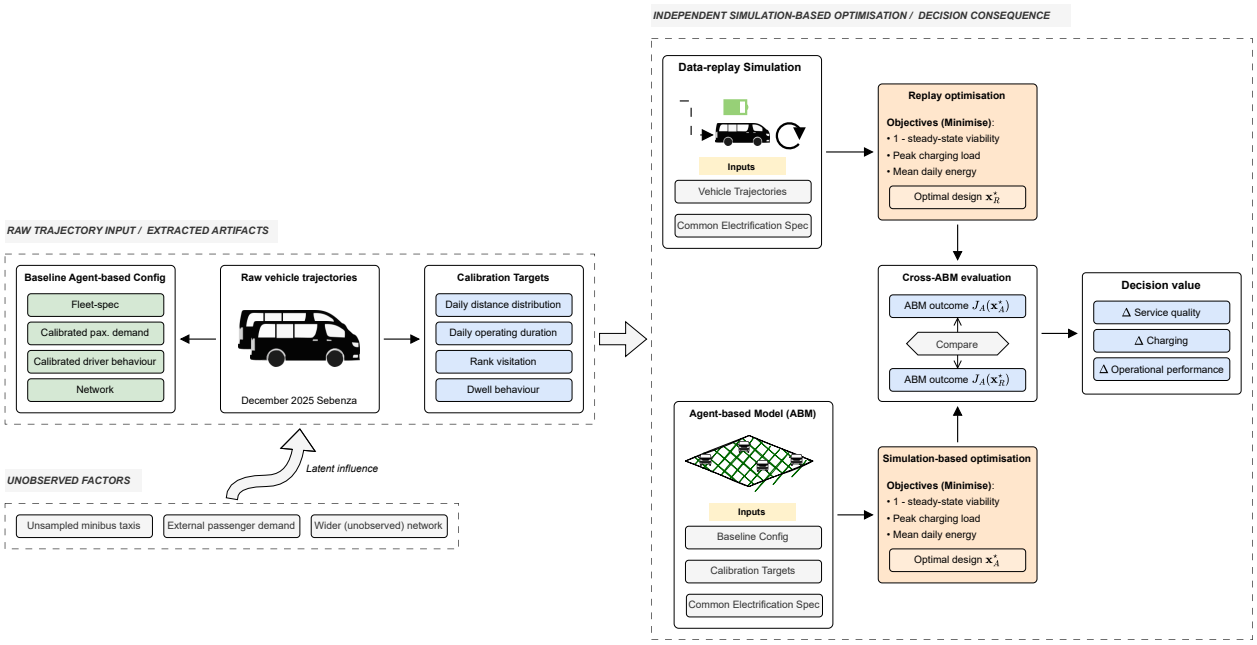


Figure 2: Overview of the study design. GPS telemetry is used to reconstruct the operational network, infer routes and identify vehicle communities. Three communities are selected using a predefined screening rule. The same observed vehicles are electrified in the data-replay model (M_{R}) and the agent-based model (M_{A}); the latter also simulates synthetic ICE vehicles operating on the same corridors. The data-replay model evaluates all candidate charging designs and selects x_R. The agent-based model searches the same design space and selects x_A. Both designs are evaluated in the agent-based model to compare their passenger-service outcomes. The three communities are then compared to assess the consistency of the results.

4.2.2. Route library and driver route sets

Typically, minibus taxis load at a terminal and depart when full [11]. A trip therefore runs between two detected anchor dwells, consisting of stop events of 10 min or more at non-home nodes. The trips of each terminal pair are grouped by the Jaccard similarity of their node sets, using a threshold of $\theta = 0.5$ []. Each group with trips on at least three vehicle-days and two dates becomes a route, represented by its medoid trip. The fleet library holds 586 observed and 503 derived reverse routes. To limit the agent's decision space, the eligible routes of a driver is the smallest set of routes that covers 80% of its trip distance on qualified routes, together with their reverse routes. Routes shorter than 5 km in the network are excluded. A vehicle with at least one eligible route becomes an agent of both models. In M_{A}, a route is one discrete unit of operation, where a driver cannot charge at a rank part-way along it.

4.2.3. Vehicle communities and pilot selection

Vehicle communities are identified from four views of observed operation: dwell time at network nodes, dwell time at formal ranks, use of network edges, and origin–destination pairs of anchor dwells. Features are weighted using term frequency–inverse document frequency (TF–IDF) []. The similarity between two vehicles is calculated as the mean cosine similarity across the views available for both vehicles.

Each vehicle is connected to its 15 nearest neighbours in a similarity graph. The Leiden algorithm partitions this graph across a range of resolution parameters, using 50 random seeds at each resolution. The seed-level partitions are combined into a consensus partition []. A resolution of 0.75 is selected, producing eight communities while providing the best stability among the resolutions that retain at least 90% of corridor use within a single community.

Candidate charging sites are identified from each community's most frequently used non-home nodes. Nodes are ranked by the number of community vehicles with dwell periods of at least 20 minutes, and the first three are retained as candidate sites in both models. Each community is then screened using the energy core of M_{R} under three charging scenarios: overnight charging only; overnight charging plus unrestricted 60 kW charging at the three candidate hubs; and depot only charging. The corresponding viable shares of vehicle-days are denoted by V_1, V_2 and V_3. For each candidate hub, the estimated connector requirement is

based on the 95th percentile of the observed daily peak number of simultaneous charging sessions in the second scenario. This screening rule was established to avoid biasing the outcomes of the data-replay agent-based comparison.

A community qualifies for pilot selection when it contains at least 15 vehicles, has a formal main hub, benefits materially from hub charging ($V_2 - V_1 \geq 15$ percentage points), and has eligible routes that represent at least 30% of the median driver's trip distance. The last condition ensures that the agent-based model represents a meaningful share of community operation. The three qualifying communities with the highest V_2 values are selected, subject to having distinct main hubs. Consequently, the results apply most directly to communities that meet these electrification-readiness criteria, rather than to all minibus-taxi communities.

4.3. Fixed-mobility data-replay

The data-replay model (M_{R}) follows the trajectory-replay approach used in earlier minibus-taxi electrification studies [8, 12]. Each observed vehicle-day is replayed using the energy-consumption and battery assumptions of the study EV [10, 9]. Trajectories, departure times and dwell periods remain fixed. A vehicle can charge when it is parked near an available charger but cannot divert from its observed movement to reach one.

A charging opportunity is defined as a stop lasting at least 10 minutes within 100 m of a candidate charging site. Vehicles from each date, called a fleet day, are replayed in chronological order, sharing the available connectors. A vehicle whose battery is below $0.9C$ joins the first-in, first-out queue and charges at the configured depot charging power (P_o) and charger efficiency (η) until either its battery reaches $0.9C$ or the observed stop ends. Since the replay uses observed dwell periods, a vehicle can charge during time spent waiting in a rank loading queue. This assumes that charging connectors are available at the loading bays. The agent-based model does not permit this as a driver must leave the loading queue to travel to a charging station. This asymmetry is retained in the comparison and favours the replay model. Additionally, every vehicle can also charge overnight, subject to the available parked time and the upper battery limit. The energy added overnight is

$$E^{\mathrm{night}} = \min\left(0.9\,C - E^{\mathrm{end}},\ \eta P_o\,\Delta t_{\mathrm{park}}\right), \qquad \Delta t_{\mathrm{park}} = t^{\mathrm{start}} + 24\,\mathrm{h} - t^{\mathrm{end}}, \tag{1}$$

where E^{end} is the battery energy at the end of driving, and t^{start} and t^{end} are the start and end of the driving window.

A vehicle-day is considered viable only if the vehicle can repeat the duty cycle without exhausting the battery. Each date is therefore simulated repeatedly, starting at $0.9C$, until the daily starting battery levels converge to a fixed point or a sustainable cycle within a tolerance of $0.001C$, all active vehicles have been resolved, or 100 iterations have been completed [12]. A vehicle that reaches zero battery energy strands and leaves the active set. Each vehicle-day is classified as viable, below reserve, stranded or unconverged. The stranded share is also reported for M_{A}, where stranding is treated as a failure condition, providing a common reliability measure for the comparison.

4.4. Decision-adaptive agent-based model

The agent-based model (M_{A}) is implemented using the the discrete-event simulation framework established by Sello and Booysen [13]. Each observed vehicle is represented by a driver agent who decides when to work, which route to operate, how long to wait for passengers, when to idle and, after electrification, when to charge. Unlike the replay model, M_{A} generates vehicle movement from these decisions.

Each simulated day represents a typical term weekday on the inferred operational network. The model specification follows the Overview, Design concepts and Details (ODD) protocol [**?**], with the full description provided in Appendix **??**. The following subsections focus on the mechanisms relevant to the comparison and on changes made relative to the earlier model.

4.4.1. Operation

When a driver is not servicing a route, the agent chooses between its eligible routes and idling using a multinomial logit model [**?** **?**]. Route utility depends on the requests waiting at the current node, the number of competing drivers, and the driver's memory of route-level demand. This memory records the number of passengers served and their passenger-distance contribution during each 30-minute interval of the day. Idling becomes more attractive during intervals that the driver remembers as having low demand.

At each stop, the driver decides whether to continue waiting according to

$$\pi = \left[1 + \exp\left(-\left(b + w_d\, d - w_o\, o\right)\right)\right]^{-1}, \tag{2}$$

where d is the remembered demand for the route, scaled by vehicle capacity, and o is the current onboard occupancy as a share of capacity. The parameters b, w_d and w_o are calibrated. The initial stop of a route represents fill-and-go loading at a rank and has separate parameters, allowing drivers to wait according to their remembered demand at the terminal.

Relative to the earlier model by Sello and Booysen [13], the stop decision now uses route-specific demand memory, rank loading is represented explicitly, and includes a last-run rule that prevents a driver from starting a route that would extend beyond its shift. A passenger boards only if the vehicle is at the request's origin, has an available seat and serves the passenger's destination. Unserviced requests expire after two hours. Travel times are drawn from windowed log-normal distributions fitted to observed network hops, while each agent's shift is based on its observed first and last movements.

4.4.2. Charging

During the electrification experiments, the observed agents become EVs, while their synthetic ICE copies remain internal combustion vehicles. Each EV driver follows a demand-aware charging policy. The vehicle always charges when its state of charge falls below $s_{\min} = 0.2$, never charges above $s_{\max} = 0.9$, and charges between these limits when

$$\frac{E(t)}{C} < s_{\min} + \left(s_{\max} - s_{\min}\right)\left(1 - \hat{h}_t\right), \tag{3}$$

where $E(t)$ is the battery energy and $\hat{h}_t \in [0, 1]$ is the normalised demand remembered by the driver for the current time interval. The policy makes charging more attractive when the driver expects charging to incur a smaller passenger-service opportunity cost.

Drivers can initiate charging only when they are not on a route. They travel to the nearest charging station and join its first-in, first-out queue. A driver waiting to load at a rank must leave the queue to charge and subsequently return to passenger service. Reserve rules reduce avoidable stranding: a driver may start a route only when the battery can support the route, the subsequent drive to the nearest station and the reserve of $0.2C$. A route is not eligible if a full battery cannot support its completion. While a vehicle is at home, it can charge at 3.7 kW. If an EV strands, it loses its onboard passengers and restarts from home with a full battery on the following day.

4.4.3. Run horizon

A cold simulation begins with empty driver memories and includes a warm-up period of 49–91 days. The warm-up ends when the seven-day means of passenger-service and route-choice measures remain close to their long-run means for three random seeds. The accepted warm-up is stored and reused for each experiment within a community. Each experiment begins from this saved warm start and runs for 14 settling days, allowing EV drivers to learn their charging behaviour. These are followed by 14 evaluation days, which provide the outcomes used for design selection and comparison.

4.5. Mixed-fleet population and passenger demand

The GPS dataset covers only a subset of the taxis operating from each hub — those fitted with Sebenza Wi-Fi routers. Thus, to represent a larger corridor population, M_{A} adds synthetic ICE copies of the observed agents. The resulting population is, on average, k times the number of agents. Each copy follows its associated agent's eligible routes, idle locations and shift pattern, but is assigned a nearby home location and a shift adjusted by a random draw with a standard deviation of 30 minutes. This construction assumes that the observed agents provide a useful representation of the wider population operating from the same hub. The electrified agents account for 2.6%–6.6% of the simulated driver population (Table 6), reflecting the small share expected in an initial pilot. The remaining synthetic vehicles retain ICE drivetrains and compete with the EVs for passenger requests.

Since GPS trajectories do not reveal passenger demand directly, the model generates a latent demand process using observed operating patterns and published passenger counts. The observed departures, routes and stops determine the structure, direction and timing of demand, while daily boardings at the main hub in

the CITP [6] determine its overall scale. Each departure has a load factor of $\lambda = 1$, except for counter-peak travel during the morning and afternoon peaks. For those departures, the load factor is set to the counter-peak-to-peak ratio of vehicle trips in CITP Table 3-18, which ranges from 0.09 to 0.39. The demand scale factor is

$$k = \frac{B}{Q\,L}, \tag{4}$$

where B is the published daily number of boardings at the main hub, Q is vehicle capacity and L is the estimated number of full agent loads from the hub per calibration day.

A share $\alpha = 0.8$ of passengers board at the route origin and alight downstream in proportion to the observed stop-event distribution. The remaining passengers travel between intermediate route nodes. Demand is generated only for origin–destination pairs that are served by at least one eligible route, follows the hourly operating profile of the agents and is sampled as a non-homogeneous Poisson process. This construction makes the simulated terminal boardings at the main hub equal the published boarding total by design.

The scale factor k is checked against an upper bound inferred from the provincial operating-licence register [14]. The Bellville estimate exceeds this bound, indicating uncertainty in the relationship between the observed agent population and the full hub population. This limitation is discussed in Section 6.3; it should not be interpreted as evidence that the synthetic population is an exact reconstruction of the entire taxi service.

4.6. Calibration and validation of the agent-based model

4.6.1. Calibration

The ICE baseline of each community is calibrated to reproduce observed operating patterns that determine an EV's daily energy requirement and opportunities for charging. Calibration targets include the 10th, 50th and 90th percentiles of daily distance, daily moving time and each agent's day-to-day coefficient of variation in daily distance. The hourly share of agents moving is also assessed, but is used as a realism check rather than a target that must be matched hour by hour. Additionally, the model must serve at least 95% of passenger requests, and median seat occupancy should approach 0.6. These are plausibility constraints and goals rather than observed calibration targets. The fit of each target is expressed as its absolute error relative to the target's estimated sampling variability, calculated from 500 bootstrap resamples of complete calibration dates.

Following Thiele et al. [15], calibration proceeds in two stages. Hand tuning first identifies a plausible region of parameter space and reveals structural gaps that cannot be resolved by parameter adjustment alone. In Khayelitsha, these gaps required a travel-time scaling adjustment and the introduction of the last-run rule. A predefined search strategy then optimises nine behavioural parameters using a constrained, two-objective tree-structured Parzen estimator (TPE) [16]. Selected configurations are validated through cold runs using three new random seeds. Finally, the accepted configuration is assessed against the held-out observation days.

4.6.2. Fidelity of the simulated movement

The movement generated by M_{A} differs from the observed movement as agents operate on a simplified network, choose from a limited set of inferred routes and stop only at network nodes. Differences between the selected charging designs could therefore arise from the representation of movement, rather than from driver adaptation alone. To assess this effect, the data-replay rules are applied to the simulated movements from five validated ICE runs of M_{A}. Each simulated home-to-home duty is treated as one vehicle-day. Distance is calculated from network travel, charging opportunities are identified from simulated dwell periods at candidate sites, and overnight energy is calculated using the same rule as in Equation (1). The first eight evaluation days of each seed are used to match the number of days replayed in M_{R}, and outcomes are averaged across the five seeds.

The resulting charging design, x_S, is the design selected by the M_{R} rule when applied to simulated rather than observed movements. Comparing x_R with x_S isolates the effect of the movement representation under the replay selection rule. Comparing x_S with x_A captures driver adaptation together with the differences between the models' service measures and selection rules. The difference in EV passenger change between x_R and x_A is decomposed as

$$\underbrace{\bar{\Delta}_{\mathrm{EV}}(x_R) - \bar{\Delta}_{\mathrm{EV}}(x_A)}_{\text{total}} = \underbrace{\bar{\Delta}_{\mathrm{EV}}(x_R) - \bar{\Delta}_{\mathrm{EV}}(x_S)}_{\text{representation}} + \underbrace{\bar{\Delta}_{\mathrm{EV}}(x_S) - \bar{\Delta}_{\mathrm{EV}}(x_A)}_{\text{behaviour}}, \tag{5}$$

where $\bar{\Delta}_{\mathrm{EV}}$ is the mean EV passenger change of Eq. (**??**). The fidelity of the movement is reported with the quantities that the replay rules use: the daily distance, the overnight time, the charging opportunities at the sites and the ceiling of the not-stranded share. No pass threshold is set.

4.7. Charging design experiment

4.7.1. Candidate sites and design space

Each charging design is represented by

$$x = (N_1, N_2, N_3, P_o), \qquad N_j \in \{0, 1, \ldots, 2\hat{N}_j\}, \qquad P \in \{22, 50\}\ \mathrm{kW}, \tag{6}$$

where N_j is the number of connectors installed at candidate hub j, and P_o is the rated power of each connector. Each hub can have between zero and $2\hat{N}_j$ connectors, where $\hat{N}_j$ is the connector estimate derived during screening. Connector power is either 22 or 50 kW. A value of $N_j = 0$ means that no charging site is installed at hub j. Site selection is therefore part of the optimisation. The resulting design spaces contain 2,645 candidates in Khayelitsha, 1,169 in Bellville and 349 in Wynberg.

4.7.2. Performance measures and selection rules

Each model evaluates designs using the measures it can represent, as summarised in Table 4. The data-replay model focuses on the share of vehicle-days that strand and the peak charging load across the candidate sites. The agent-based model reports these measures and also represents passenger demand, waiting times, expired requests and the passenger throughput of the EV agents.

Table 4: Model-native measures of a charging design. All measures use the evaluation days of M_{A} or the replayed term weekdays of M_{R}. A dash shows a measure that the model cannot represent.

Measure	**Definition**	M_{R}	M_{A}
Stranded share	EV vehicle-days whose battery level reaches 0	Front and selection	Reported (must be 0)
Below-reserve share	EV vehicle-days whose lowest level is below $0.2\,C$	Reported	Reported
Peak site load (kW)	Peak of the summed grid power of all site sessions	Front and selection	Search and selection
Site and home energy (kWh a day)	Grid energy at the sites and at the homes	Reported	Reported
Expired share	Expired requests over all requests (EVs and ICE vehicles)	–	Search and selection
Mean passenger wait (min)	Time from the request to the boarding of served requests	–	Search
EV passengers carried	Passengers that the EV agents carry	–	Search
EV passenger change	EV passengers against the same agents as ICE vehicles	–	Selection
EV passenger loss ℓ	EV passenger change corrected for the time away from home	–	Reported

In the replay model, the stranded share is the fraction of conclusive vehicle-days that end in stranding. The peak site load is the maximum simultaneous charging load summed across all sites. These measures define the replay model's design-selection objective space. In the agent-based experiments, every EV run is paired with an ICE counterfactual using the same configuration, random seed and warm start. In the counterfactual, the observed agents operate as ICE vehicles rather than EVs. Pairing these runs uses common random numbers to reduce simulation noise in the comparison.

The EV passenger change and the EV passenger loss are

$$\Delta_{\mathrm{EV}} = \frac{M_{\mathrm{EV}}^{\text{EV run}}}{M_{\mathrm{EV}}^{\text{ICE run}}} - 1, \qquad \ell = 1 - \frac{R^{\text{EV run}}}{R^{\text{ICE run}}}, \qquad R = \frac{M_{\mathrm{EV}}/M}{Y_{\mathrm{EV}}/Y}, \tag{7}$$

where M is the number of passengers carried by all drivers on the evaluation days, Y is the time that all drivers spend away from home, and the subscript EV denotes the agents. ℓ corrects Δ_{EV} for a change in the agents' time on the road. It is therefore a measure of passenger throughput relative to operating time, not a measure of passengers lost by the transport system as a whole.

The replay model is deterministic, allowing every candidate design to be evaluated. Its Pareto front is therefore exact within the specified design space. Because replayed vehicles cannot divert to a charging station, even unlimited connectors cannot make every vehicle-day viable. A design with 99 connectors at each candidate site is used to establish the attainable ceiling for the share of vehicle-days that avoid stranding. The replay candidates must reach at least 95% of this ceiling. Among those candidates, x_R is selected by minimising peak site load and then installed power. The selected design is used as the common reference for the first research question.

The agent-based model is more computationally expensive, so it cannot feasibly evaluate every candidate design. Instead, a multi-objective TPE search evaluates 40 trials using one seed per trial, starting from x_R. The search minimises expired passenger requests, mean passenger waiting time and peak site load, while maximising the passengers carried by the EV agents. The resulting candidates and x_R are then evaluated on five new seeds with their paired ICE counterfactuals. The final agent-based design, x_A, is chosen using a one-standard-error rule. To avoid selecting a design solely because it carries marginally more passengers when that apparent advantage may be due to simulation randomness, a candidate must achieve an EV passenger change no more than one standard error below the best validated value, while its mean expired-request share must be within one percentage point of the minimum. Among qualifying candidates, the selected design has the lowest peak site load and then the lowest installed power. No design is selected manually. Any validated EV run that ends in stranding invalidates the selected result.

4.7.3. Cross-evaluation, sensitivity and synthesis

The consequences of selecting infrastructure under fixed-mobility assumptions are measured by evaluating x_R and x_A in M_{A} using the same five validation seeds. For each outcome, the paired difference is calculated between the two designs on each seed and then averaged. The standard error of the mean paired difference quantifies the uncertainty across the five seeds. Since both designs use the same seed and warm start, the paired comparison reduces variation arising from different random simulation conditions. Differences within two standard errors of zero are treated as indistinguishable at the resolution of this experiment.

Changes in expired requests are also examined. If EV passenger throughput falls while the overall expired-request share remains unchanged, the result is consistent with passengers being served by competing ICE taxis rather than being lost from the system. However, this inference should be interpreted alongside the passenger-service measures actually reported by the model.

Robustness is assessed by evaluating x_R and x_A, without reselecting either design, under six alternative conditions: energy consumption 15% below and above the baseline; charging-policy minimums of 0.1 and 0.35; and fleet scale factors k equal to 0.5 and 1.5 times their baseline values. Each case is evaluated using three seeds, with a corresponding ICE counterfactual where the changed condition affects ICE operation.

For each community, the analysis records three results: the difference in installed power between x_A and x_R, the EV passenger-service cost of using x_R rather than x_A, and the EV passenger change achieved by x_A relative to the ICE counterfactual. A result is considered robust when its direction is unchanged across all tested cases, and consistent across communities when the same direction occurs in all three. These experiments assess the robustness of the selected designs' outcomes. They do not test whether the designs themselves would remain optimal under the alternative conditions.

5. Results

5.1. Selected communities and model validation

5.1.1. Communities

The community-detection procedure identifies eight vehicle communities (Table 5). Hub charging improves viability in every community, but the attainable viable share remains limited because the fleet's median daily distance of 190 km exceeds the EV's approximate 130 km operating range. The predefined screening rule selects Bellville ($V_2 = 73.6\%$), Wynberg ($V_2 = 50.3\%$) and Khayelitsha ($V_2 = 36.9\%$). Three communities fail at least one eligibility criterion, while two eligible communities rank below the selected three.

Table 5: Electrification screening of each community and the reason for each exclusion. V_1, V_2 and V_3 are the shares of the screened vehicle-days that are viable with overnight charging only, with overnight and hub charging at the first three hubs, and with hub charging only. Binding is $V_2 - V_1$ in percentage points. A main hub is formal when it is a rank, a station or a terminus.

Community	Vehicles	Main hub	V_1 (%)	V_2 (%)	V_3 (%)	Binding (pp)	Selected	Reason
C1 Khayelitsha	101	Rank	14.4	36.9	5.9	22.5	Yes	–
C2	65	Terminus	15.1	34.4	2.6	19.3	No	A higher-ranked candidate was selected
C3	51	Rank	15.5	34.4	8.6	18.9	No	A higher-ranked candidate was selected
C4 Bellville	44	Rank	23.9	73.6	2.5	49.7	Yes	–
C5 Wynberg	26	Terminus	21.9	50.3	4.7	28.4	Yes	–
C6	17	Terminus	31.5	51.4	3.6	19.8	No	Routes cover too little of the driving
C7	17	Terminus	55.0	70.6	15.6	15.6	No	Routes cover too little of the driving
C8	14	Rank	45.5	71.6	9.1	26.1	No	Fewer vehicles than the minimum

The selected communities differ in how well their operation is represented by the inferred route sets and network. Their median route coverage ranges from 36.9% to 62.9% of trip distance, while the share of driving outside the operational network ranges from 18.6% to 41.0% (Table 6). These differences are important when interpreting the agent-based results, because its drivers are restricted to the inferred network and their eligible routes.

Table 6: The selected communities, their data and their configured models. Agents are the EV drivers and their synthetic ICE copies in M_{A}, with the scale factor k. Routes are the distinct configured routes of the drivers. Valid: the valid share of the community's vehicle-days. Off net.: the share of the driving outside the network. Route cov.: the median share of a vehicle's trip distance on its eligible routes, over the 93, 44 and 25 vehicles with an eligible route before routes shorter than 5 km were excluded.

Community	Vehicles	Qualified Vehicles	Agents	k	Routes	Requests a day	Valid (%)	Off net. (%)	Route cov. (%)
Khayelitsha	101	89	1,358	15.21	395	78,561	67.8	32.5	36.9
Bellville	44	44	1,673	38.03	259	130,881	68.6	18.6	62.9
Wynberg	26	20	686	34.30	114	43,172	61.6	41.0	41.0

5.1.2. Calibration

The calibrated ICE baselines reproduce the central parts of the observed duty-cycle distributions that determine EV energy use and charging opportunities (Table 7). The differences between observed and simulated median daily distance are 2.3, 12.9 and 1.4 km in Khayelitsha, Bellville and Wynberg, respectively. The corresponding differences in median daily moving time are 0.3, 0.2 and 0.5 hours. In all three communities, the model serves at least 95.2% of passenger requests.

The main remaining discrepancies occur in the distribution tails. In Wynberg, the 10th percentile of simulated daily distance is 91 km, compared with 113 km in the observations. The simulated agents also vary less from day to day than the observed vehicles: the 90th percentile of the distance coefficient of variation is 0.26–0.35 in the simulation, compared with 0.37–0.75 in the observations. The simulated hourly activity profiles also have lower peaks and relatively more activity around midday. Despite these differences, the accepted configurations meet the predefined hold-out criteria. The hold-out gaps are 0.83, 0.58 and 0.65 in Khayelitsha, Bellville and Wynberg, respectively, all below their corresponding acceptance limits of 1.31, 1.69 and 1.75. This supports the use of the calibrated models for the planned comparisons, while the residual differences in activity patterns remain relevant to their interpretation.

5.1.3. Agreement between observed and simulated movement

Table 8 compares observed vehicle movements with the movements generated by the calibrated ICE baselines of M_{A}, using the quantities required by the replay rules. Daily distance is broadly consistent

Table 7: Duty-cycle fit of the baseline agent-based model of each community. Obs.: the calibration target from the agent-days of the observed agents on the calibration days. M_{A}: the run of the accepted configuration from its warm start. M_{R} replays the observed movement, so it reproduces the observed values by construction and is not shown. The served share (constraint: at least 0.95) and the seat occupancy (goal: up to 0.6) are plausibility checks with no observed value.

	Khayelitsha		Bellville		Wynberg	
Measure	Obs.	M_{A}	Obs.	M_{A}	Obs.	M_{A}
Daily distance, P10 (km)	126.4	120.2	110.3	114.0	113.1	91.0
Daily distance, P50 (km)	212.6	210.3	150.5	163.4	200.5	201.9
Daily distance, P90 (km)	323.4	317.9	226.7	231.6	298.7	318.4
Moving time, P10 (h)	5.20	4.82	3.74	3.65	4.34	2.99
Moving time, P50 (h)	8.27	8.58	5.20	5.05	6.04	6.58
Moving time, P90 (h)	11.47	12.21	7.52	6.83	9.58	9.55
Daily distance CV, P10	0.04	0.13	0.09	0.13	0.08	0.14
Daily distance CV, P50	0.15	0.20	0.14	0.17	0.14	0.20
Daily distance CV, P90	0.75	0.34	0.37	0.26	0.73	0.35
Served requests (share)	–	0.958	–	0.952	–	0.967
Seat occupancy on route runs	–	0.39	–	0.57	–	0.44

between observed and simulated movements in all three communities, suggesting reasonable agreement in total daily energy requirements under the assumed consumption rate.

However, two differences remain. First, simulated overnight periods are 1.5–2.5 hours longer at the median than observed periods. This difference provides the simulated EVs with an additional 5–8 kWh of potential home-charging energy per day, equivalent to 6%–15% of the median daily energy requirement. Second, the frequency of charging opportunities at candidate sites differs across communities. In Khayelitsha, the simulated agents stop at candidate sites about half as often as the observed vehicles, reducing the maximum attainable share of viable vehicle-days from 45.1% to 31.0%. In Bellville, simulated agents stop at candidate sites more often than observed vehicles. In Wynberg, the observed and simulated movements provide similar numbers of site opportunities. Figure 3 presents the full distributions.

Table 8: Observed against simulated movement, measured with the quantities of the replay rules. Obs.: the replayed vehicle-days of the agents on the eight term weekdays. Sim.: the home-to-home duties of the agents in the calibrated ICE runs of M_{A} (first eight evaluation days of five seeds). Distances and times are P50 [P10–P90]. A site opportunity is a stop of 10 min or more within 100 m of a candidate site. The battery window is $0.7\,C$. The ceiling is the highest not-stranded share that M_{R} reaches with unlimited connectors at the three sites.

	Khayelitsha		Bellville		Wynberg	
Measure	Obs.	Sim.	Obs.	Sim.	Obs.	Sim.
Vehicle-days	463	3,424	234	1,697	99	773
Daily distance (km)	208 [127–309]	205 [116–321]	145 [109–220]	162 [105–232]	185 [119–288]	198 [98–320]
Overnight time (h)	6.9 [2.6–10.0]	8.4 [4.1–13.0]	5.7 [2.8–9.6]	8.2 [4.4–10.6]	5.7 [2.7–9.0]	7.8 [2.3–11.2]
Site opportunities per vehicle-day	2.44	1.22	4.08	6.75	1.99	1.70
Days without a site opportunity (%)	28.5	34.1	8.1	0.5	33.3	32.1
Days that need more than the window (%)	89.2	86.7	65.4	75.8	81.8	77.4
Ceiling of the not-stranded share (%)	45.1	31.0	80.8	84.6	53.1	47.5

5.2. Electrification outcomes under a common design (RQ1)

Under the reference design x_R, the two models predict substantially different electrification outcomes (Table 9). In the data-replay model, 57.0%, 22.6% and 49.0% of vehicle-days strand in Khayelitsha, Bellville, and Wynberg, respectively. These failures largely reflect the mismatch between daily energy requirements and the charging opportunities available along the fixed trajectories. Median daily energy needs are 79, 55 and 70 kWh, whereas overnight charging at 3.7 kW supplies approximately 23, 19 and 19 kWh. Consequently, 94%–97% of vehicle-days require site charging to sustain repeated operation. Yet 29%, 8% and 33% of days, respectively, contain no qualifying stop near a candidate charging site. Even with unlimited connectors, the replay model therefore predicts that 55%, 19% and 47% of vehicle-days would still strand in the three communities (Table 8).

In the agent-based model, no validated EV run strands because drivers can travel to charging stations. This improved ability to obtain energy comes with a passenger-service cost. Under the same reference design,

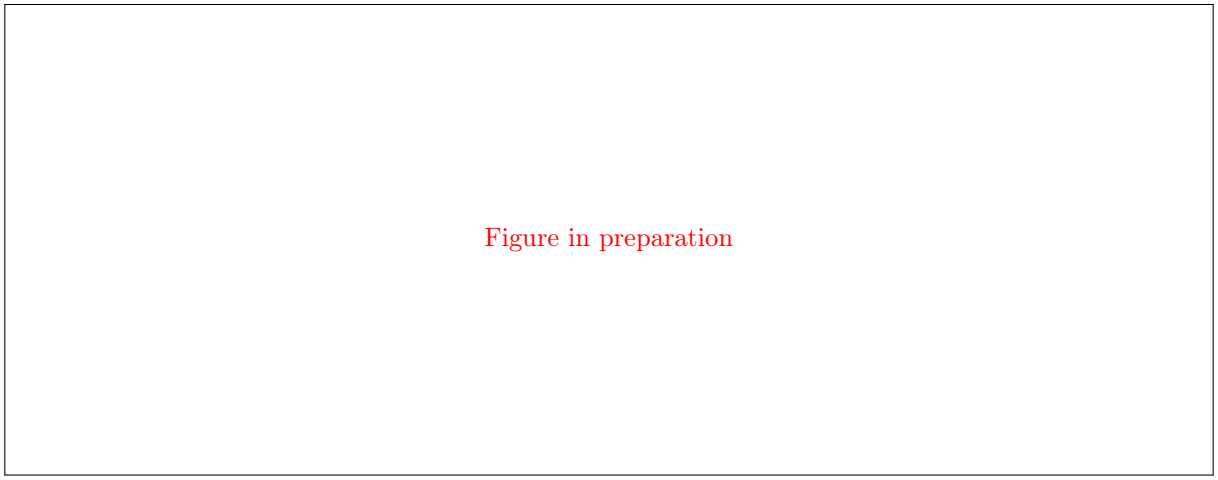


Figure 3: Observed against simulated movement in each community: daily distance, overnight time and charging opportunities at the candidate sites.

site energy use is 2.8, 1.8 and 4.0 times the replay values in Khayelitsha, Bellville and Wynberg, respectively. The model also records 2.4, 1.5 and 3.8 times as many site charging sessions. The electrified agents nevertheless carry 18.5%, 6.0% and 23.8% fewer passengers than in their corresponding ICE counterfactuals.

For this common design, the models predict the same peak site load because all installed connectors are used at the peak. The replay model reports lower site energy partly because stranded vehicles stop charging and leave the active fleet. Figure 4 illustrates the differences in charging patterns, battery levels and the causes of replay failure.

Table 9: Outcomes of the selected M_{R} design of each community in both models. M_{R}: the replayed term weekdays; M_{A}: the mean over five validation seeds of 14 evaluation days each. The below-reserve share of M_{R} counts the vehicle-days below $0.2\,C$; M_{A} reports it with another definition, so its cell is empty. A dash shows a measure that the model cannot represent.

	Khayelitsha		Bellville		Wynberg	
Measure	M_{R}	M_{A}	M_{R}	M_{A}	M_{R}	M_{A}
Stranded EV-days (share)	0.570	0.000	0.226	0.000	0.490	0.000
EV-days below the reserve (share)	0.646	–	0.274	–	0.602	–
Peak site load (kW)	250	250	200	200	100	100
Site energy (kWh a day)	1,368	3,797	1,020	1,819	299	1,196
Home energy (kWh a day)	356	2,104	323	1,089	129	445
Site sessions a day	99.1	237.9	83.1	124.3	16.4	62.0
Expired requests (share)	–	0.041	–	0.049	–	0.033
Mean passenger wait (min)	–	2.1	–	4.5	–	16.8
EV passenger change against ICE	–	−0.185	–	−0.060	–	−0.238
EV passenger loss ℓ	–	0.192	–	0.053	–	0.252

5.3. Selected charging infrastructure (RQ2)

The models select different charging designs in all three communities, and the replay-selected design x_R is not a qualifying candidate under the agent-based selection rule in any community (Table 10 and Figure 5). The scale of the difference varies by community. In Khayelitsha, the agent-based model selects 1,000 kW of installed power, compared with 250 kW in the replay model. The selected design also has a higher peak site load: 800 kW rather than 250 kW. The agent-based search shows that additional connectors initially produce substantial passenger-service gains. The EV passenger loss falls by 46 percentage points between 22 and 250 kW of installed power, and by a further 14 points between 250 and 670 kW. These gains arise because drivers can travel to the charging sites. Under the replay model, the same additional capacity has little effect on stranding: the stranded share falls only from 57.0% under x_R to 54.9% under x_A, since observed vehicles cannot divert to use the new capacity. Above 670 kW, the additional passenger-service gain is only

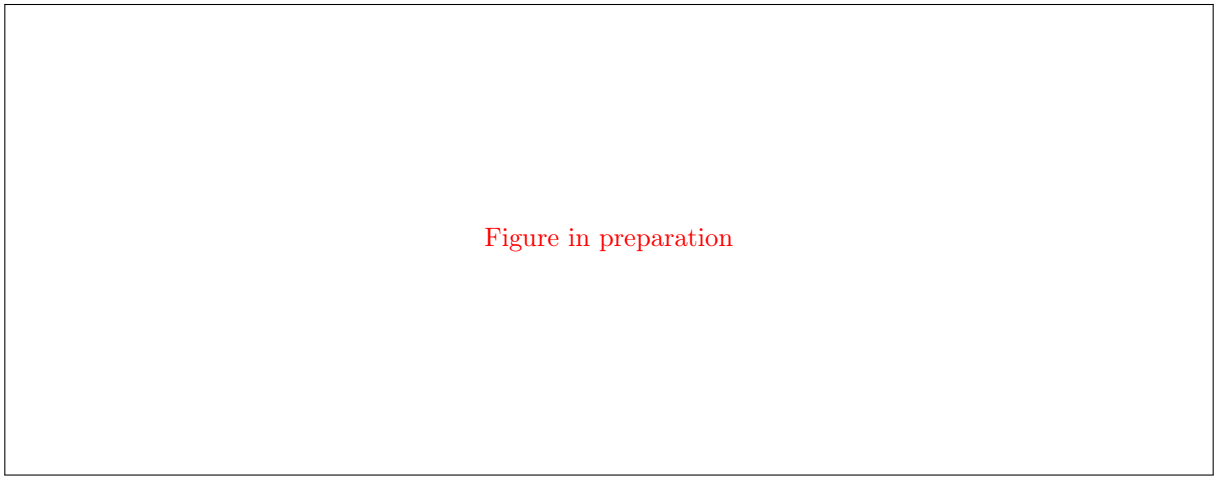


Figure 4: Charging under the reference design in both models: site load over the day, the causes of the replay failures, and the battery level at the start of the day.

1.4 percentage points. Nevertheless, the selection rule chooses the 800 kW peak-load design because it has no direct cost penalty for installed capacity beyond the stated load-based criteria.

In Bellville, both models select 200 kW of installed power, but they allocate the connectors differently. The replay model assigns one connector to the first hub and three to the second, whereas the agent-based model assigns all four to the second hub. In Wynberg, the agent-based model selects 150 kW instead of 100 kW, with a different allocation between the first and third hubs. Above 150 kW, further connector additions do not reduce the EV passenger loss. The design differences are therefore not solely a question of total capacity. In Bellville and Wynberg, the location of the connectors matters even when the total installed power is similar.

Table 10: Design that each model selects in each community: the connectors N_1, N_2 and N_3 at the first three hubs, the power P of a connector, the installed power and the peak site load. A design ID $n\,N_1$-N_2-N_3_$p\,P$ gives the same values.

Community	Model	Design	N_1	N_2	N_3	P (kW)	Installed (kW)	Peak (kW)
Khayelitsha	M_{R}	n03-01-01_p50	3	1	1	50	250	250
	M_{A}	n08-07-05_p50	8	7	5	50	1,000	800
Bellville	M_{R}	n01-03-00_p50	1	3	0	50	200	200
	M_{A}	n00-04-00_p50	0	4	0	50	200	200
Wynberg	M_{R}	n01-01-00_p50	1	1	0	50	100	100
	M_{A}	n02-00-01_p50	2	0	1	50	150	150

5.3.1. Representation and behaviour effects

The simplified movement representation does not explain most of the design difference in Khayelitsha (Table 11). When the replay selection rule is applied to simulated ICE movements, the resulting design x_S differs from x_R by only one connector in each community. In Khayelitsha, the remaining 800 kW difference between x_S and x_A is 16 times the 50 kW difference between x_R and x_S. The result is therefore driven mainly by the operational responses and selection criteria represented in M_{A}, rather than by the movement representation alone.

In Bellville and Wynberg, the difference in installed power between x_R and x_A is one connector or less. The principal change is the allocation of connectors between hubs. These results suggest that replay-based planning is most likely to understate infrastructure requirements where observed vehicles rarely stop at candidate sites but can gain passenger service by travelling to charge. However, the decomposition also combines behavioural effects with differences in the models' service measures and selection rules; it should not be interpreted as a pure causal estimate of behaviour alone.

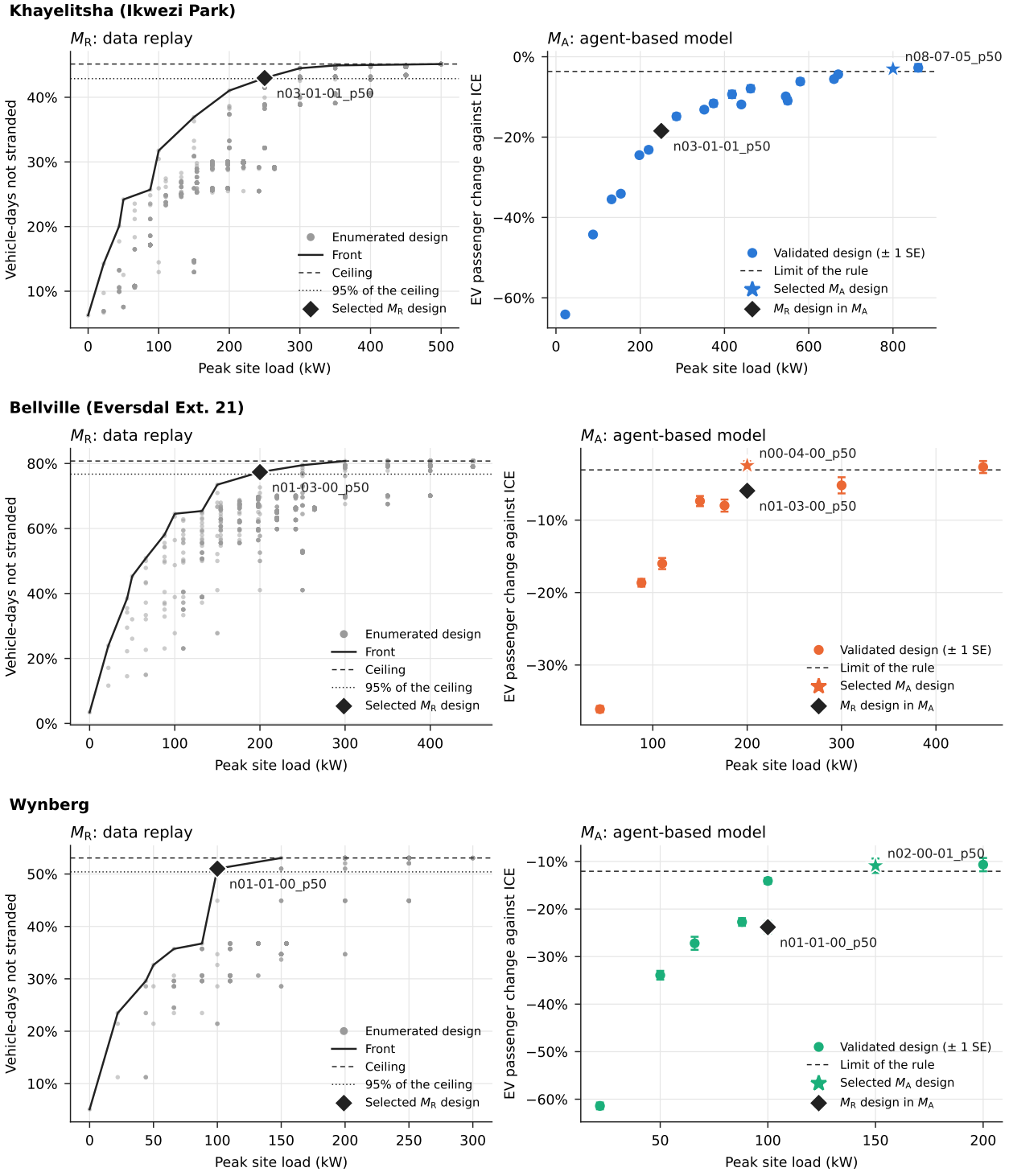


Figure 5: Design space of each community in both models, each in its model-native objectives: (a) Khayelitsha, (b) Bellville, (c) Wynberg. Left, M_R: every enumerated design (grey), the front (line), the ceiling of the not-stranded share (dashed) and 95% of it (dotted), and the selected M_R design (diamond). Right, M_A: the validated designs (mean ± one standard error over five seeds), the limit of the one-standard-error rule (dashed), the selected M_A design (star) and the M_R design evaluated in M_A (diamond).

Table 11: Representation and behaviour effects. x_S is the design that the M_{R} rule selects on the simulated ICE movement of M_{A}. The step $x_R \to x_S$ isolates the representation effect; $x_S \to x_A$ contains the behaviour effect and the change of service measure and selection rule. Designs are N_1-N_2-N_3 connectors of 50 kW, with the installed power in kW. ΔkW: the change of installed power; moved: half the summed absolute change of the connectors per site. Cost: the EV passenger change in M_{A} of the first design minus that of the second, with its standard error over five paired seeds; the two parts sum to the cost in Table 12.

				Representation ($x_R \to x_S$)		Behaviour ($x_S \to x_A$)	
Community	x_R	x_S	x_A	ΔkW (moved)	Cost (SE)	ΔkW (moved)	Cost (SE)
Khayelitsha	3-1-1 (250)	3-1-0 (200)	8-7-5 (1,000)	−50 (0.5)	+0.065 (0.003)	+800 (8.0)	−0.220 (0.008)
Bellville	1-3-0 (200)	2-2-1 (250)	0-4-0 (200)	+50 (1.5)	−0.005 (0.009)	−50 (2.5)	−0.029 (0.013)
Wynberg	1-1-0 (100)	0-1-1 (100)	2-0-1 (150)	0 (1.0)	−0.006 (0.003)	+50 (1.5)	−0.124 (0.005)

5.4. Cost of the replay design and the EV passenger loss (RQ3)

When evaluated in M_{A}, the replay-selected design x_R results in an EV passenger change that is 15.4, 3.5 and 12.9 percentage points lower than the agent-based design x_A in Khayelitsha, Bellville and Wynberg, respectively (Table 12). Each difference exceeds two standard errors in magnitude. In Khayelitsha, the difference is primarily associated with the amount of installed charging capacity. In Bellville and Wynberg, connector location explains much of the difference.

The effect of location is apparent even when installed power is held constant. In Wynberg, the 100 kW allocation selected by the agent-based model performs better than the replay allocation, with a difference of 9.7 percentage points in EV passenger change. In Bellville, reallocating the same 200 kW of capacity produces a 3.5-percentage-point improvement. The representation component of the decomposition is comparatively small or acts in the opposite direction to the total effect (Table 11). In Khayelitsha, x_S has one fewer connector than x_R and performs worse in passenger terms. The total difference between x_R and x_A therefore understates the improvement associated with the behaviour-and-selection component.

The EV passenger loss is not accompanied by a comparable increase in expired passenger requests. Under the two designs, the expired-request share is effectively unchanged in Khayelitsha and Bellville, and differs by only 0.1 percentage points in Wynberg. Mean passenger waiting time also changes by less than 10 seconds in each community. These results are consistent with ICE taxis absorbing much of the passenger demand that the EVs do not serve while charging.

Even under the agent-based design x_A, EV passenger throughput remains below the ICE counterfactual. The time-adjusted EV passenger loss ℓ is 3.2% in Khayelitsha, 2.1% in Bellville and 11.8% in Wynberg. These remaining losses represent the operational cost of electrification under the selected charging infrastructure and the modelled mixed-fleet conditions. No validated EV run strands under either design. Figure 6 shows how passenger losses vary over the day and how they relate to charging activity.

Table 12: The selected M_{R} design x_R against the selected M_{A} design x_A, both evaluated in M_{A}: means over five validation seeds, and the paired difference (x_R minus x_A) with its standard error. No EV-day strands in any run.

	Khayelitsha			Bellville			Wynberg		
Measure	x_R	x_A	Diff. (SE)	x_R	x_A	Diff. (SE)	x_R	x_A	Diff. (SE)
EV passenger change	−0.185	−0.030	−0.154 (0.008)	−0.060	−0.025	−0.035 (0.011)	−0.238	−0.109	−0.129 (0.006)
EV passenger loss ℓ	0.192	0.032	0.160 (0.007)	0.053	0.021	0.032 (0.006)	0.252	0.118	0.134 (0.005)
Expired requests (share)	0.041	0.041	0.000 (0.000)	0.049	0.049	0.000 (0.000)	0.033	0.034	0.000 (0.001)
Mean passenger wait (s)	125.4	122.9	2.6 (0.7)	268.1	270.9	−2.8 (3.0)	1,006.0	1,015.3	−9.4 (11.9)
Peak site load (kW)	250	800	−550 (15.8)	200	200	0 (0.0)	100	150	−50 (0.0)
Site energy (kWh a day)	3,797	5,012	−1,215 (34.6)	1,819	1,925	−105 (25.6)	1,196	1,444	−247 (23.1)
Site sessions a day	237.9	343.6	−105.7 (1.8)	124.3	131.2	−6.8 (2.8)	62.0	86.0	−24.0 (1.8)

5.5. Robustness, consistency and computational cost (RQ4)

Across all six sensitivity cases, the replay-selected design x_R produces a larger EV passenger loss than the agent-based design x_A in all three communities (Figure 7 and Table 13). The difference ranges from 8.8 to 20.2 percentage points in Khayelitsha, 0.9 to 3.0 points in Bellville, and 4.6 to 18.4 points in Wynberg. Some Bellville differences are small relative to the variability across three simulation seeds, so the direction is more consistent than the magnitude.

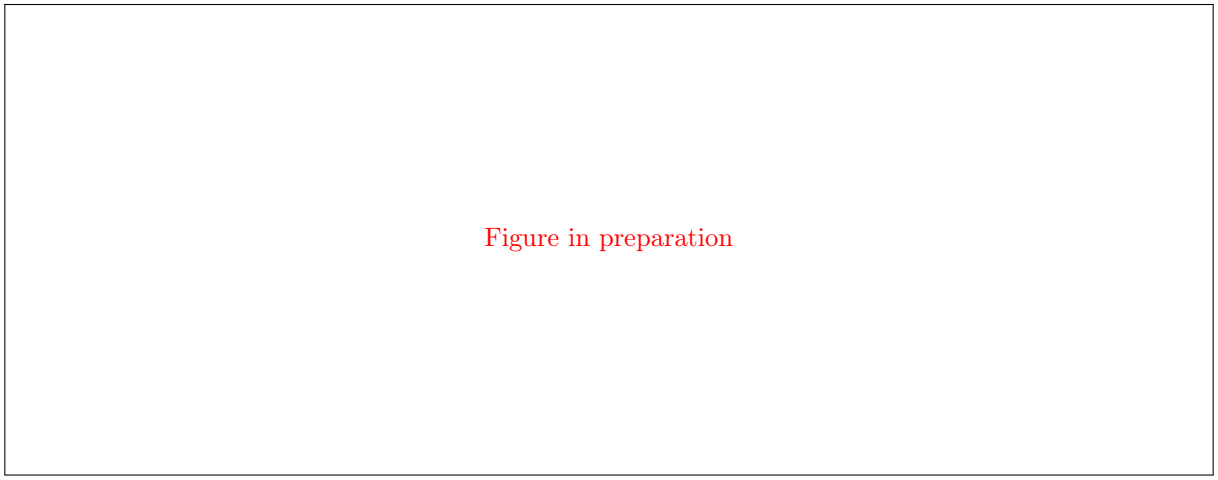


Figure 6: Passenger loss of the EVs over the day and the time that they spend on charging, under the replay design and the agent-based design.

Higher energy consumption generally widens the gap between the designs, while lower consumption narrows it. The additional overnight charging time in simulated movement gives EVs extra potential energy relative to observed movement. However, the advantage of the agent-based design persists even under the sensitivity case with 15% lower energy consumption. Across all tested cases, only two EV-days strand, both in Bellville when the charging-policy minimum is reduced to $s_{\min} = 0.1$.

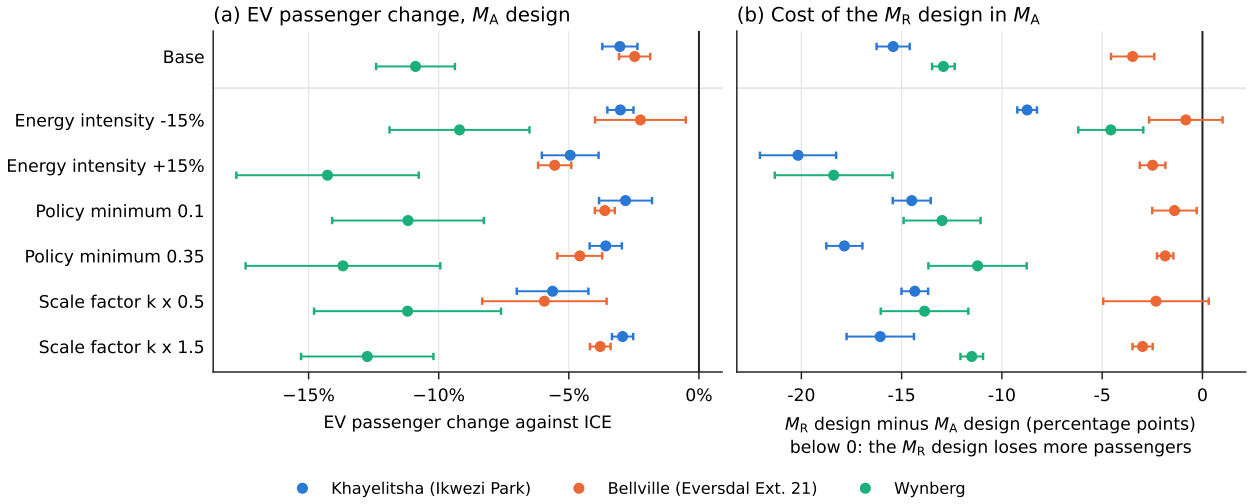


Figure 7: Sensitivity of the EV passenger change in M_A in each community (colour). (a) EV passenger change against internal-combustion operation of the selected M_A design. (b) EV passenger change of the selected M_R design minus that of the selected M_A design, both evaluated in M_A, in percentage points; below zero, the M_R design loses more passengers. Mean ± one standard error over three seeds in each case and five in the base case.

The cross-community synthesis (Table 14) shows that the passenger-service consequence of using x_R instead of x_A has the same direction in all three communities: the replay-selected design carries fewer passengers in M_A. Installed power is higher under x_A in Khayelitsha and Wynberg and equal in Bellville. Only the Khayelitsha difference is substantially larger than the movement-representation effect. Thus, the direction of the passenger-service result is consistent across the selected communities, but the magnitude and form of the infrastructure change depend on their operating patterns.

Table 13: EV passenger change against internal-combustion operation of the selected M_{A} and M_{R} designs, both evaluated in M_{A}, in each sensitivity case. Means over three seeds; the base case uses the five validation seeds.

	Khayelitsha		Bellville		Wynberg	
Case	x_A	x_R	x_A	x_R	x_A	x_R
Base	−0.030	−0.185	−0.025	−0.060	−0.109	−0.238
Energy intensity −15%	−0.030	−0.118	−0.022	−0.031	−0.092	−0.138
Energy intensity +15%	−0.049	−0.251	−0.055	−0.080	−0.143	−0.327
Policy minimum 0.1	−0.028	−0.173	−0.036	−0.050	−0.112	−0.242
Policy minimum 0.35	−0.036	−0.214	−0.046	−0.064	−0.137	−0.249
Scale factor $k \times 0.5$	−0.056	−0.200	−0.059	−0.083	−0.112	−0.251
Scale factor $k \times 1.5$	−0.029	−0.190	−0.038	−0.068	−0.127	−0.243

Table 14: Direction and size of each difference between the models in each community: whether the two models select the same design, the installed power of the M_{A} design minus that of the M_{R} design, and the EV passenger change of the M_{R} design minus that of the M_{A} design in M_{A} with its standard error (equal within two standard errors). Three communities do not give a statistical generalisation.

	Selected design			Installed power		EV passenger change	
Community	M_{R}	M_{A}	Same	Difference (kW)	Direction	Difference (SE)	Direction
Khayelitsha	n03-01-01_p50	n08-07-05_p50	No	+750	Higher	−0.154 (0.008)	Lower
Bellville	n01-03-00_p50	n00-04-00_p50	No	0	Equal	−0.035 (0.011)	Lower
Wynberg	n01-01-00_p50	n02-00-01_p50	No	+50	Higher	−0.129 (0.006)	Lower

6. Discussion

TODO: Complete discussion.

6.1. When does driver agency matter?

The results show that driver agency matters when the observed movement does not provide sufficient charging opportunities to sustain repeated operation, hinders the use of additional installed capacity, or does not reflect the operational effects of charging on passenger service. For Khayelitsha, driver adaption changes both the amount of power and passenger service. The selected installed power rises from 250 kW to 1,000 kW, and the EV passenger loss falls from 19.2% to 3.2%. In Bellville, the selected installed power remains at 200 kW, but the connector allocation changes and the EV passenger loss falls from 5.3% to 2.1%. In Wynberg, both the installed power and connector allocation change, and the EV passenger loss falls from 25.2% to 11.8%. Subsequently, the results do not establish that the agent-based model is universally preferable but rather that it is more appropriate when designs can be reliably ascertained from calibrated operating patterns, inferred networks, synthetic fleet and latent passenger-demand assumptions.

Within the context of electrifying Cape Town's paratransit, a particularly important distinction is that the agent-based model can represent the operational effects of charging on passenger service, while the replay model cannot. The agent-based model can capture how drivers adapt to charging opportunities and how this adaptation affects passenger throughput, which is crucial for evaluating the effectiveness of different charging designs. In contrast, the replay model assumes that drivers will follow their observed patterns without adaptation, potentially leading to suboptimal design choices.

6.2. Implications for electrification pilots

This study's findings show three key implications for electrification pilots. First, the results show that electrification pilots should consider the potential for driver adaptation when designing charging infrastructure for paratransit. Charging infrastructure should not be sized using range and observed trajectories alone when vehicles can change their movement patterns to access charging opportunities. Instead, pilots should account for the operational effects of charging on passenger service and the potential for drivers to adapt their routes and passenger seeking behaviour. Second, the results show that site allocation matters. For example, Bellville shows that two designs with the same installed power but different connector allocations can lead to different passenger service outcomes. Third, the consequences of charging extend beyond the electrified vehicles themselves. The results show that the EV passenger loss is not accompanied by a comparable increase in

expired passenger requests, suggesting that ICE taxis absorb much of the passenger demand that the EVs do not serve while charging. This highlights the potential for electrification to undermine the ability of drivers to remain competitive in a mixed fleet, and the need to consider the broader transportation ecosystem when designing charging infrastructure to mitigate these effects.

These findings support evaluating candidate designs using both energy-feasibility measures and measures of passenger service. They do not, by themselves, establish which sites are technically or commercially feasible in practice. The study's candidate hubs, connector design space and charging assumptions remain a model-based representation of a possible pilot.

6.3. Limitations

Important limitations identified in this study include the selection of communities, the use of observed and synthetic fleets, the representation of networks and routes, calibration and behavioural assumptions, comparison between models, and temporal generalisability. The screening rule selects communities partly on their replay-based electrification outcomes, which may limit the applicability of findings to other communities. The electrified agents come from the observed telemetry subset, while the rest of the simulated fleet is constructed synthetically, which may affect the representativeness of the results. The inferred network does not capture all observed driving, and the eligible route sets cover only part of each agent's observed trip distance, limiting the scope of movements that can be reproduced. Calibration reproduces central duty-cycle statistics reasonably well but underestimates day-to-day variability and does not fully reproduce the hourly activity profile. The charging policy and latent passenger demand also affect outcomes. The two models have different representations of mobility, passenger service, and charging opportunities, which means that results should be interpreted as a comparison of modelling approaches rather than a controlled estimate of behaviour alone. Finally, three selected communities and a short December observation window cannot establish statistical generalisability across South African minibus-taxi operations or across different seasons.

6.4. Future work

Future research should address the limitations identified in this study. Testing more communities and observation periods would help establish statistical generalisability across South African minibus-taxi operations and different seasons. Improving the representation of route and network coverage would allow for a more accurate assessment of charging infrastructure needs. At the time of writing, no electric minibus-taxi pilot has been implemented in South Africa, so the study relies on model-based representations of possible pilots. However, obtaining direct observations of charging and driver decisions would provide valuable data for calibrating models and validating assumptions. Incorporating constraints from the electricity distribution network would ensure that proposed charging designs are feasible within the existing infrastructure. Additionally, future work could compare the computational and planning value of replay and agent-based simulation across a wider range of candidate pilots, providing further insights into the most effective modelling approaches for electrification planning.

7. Conclusion

In conclusion, this study demonstrates that representing driver adaptation changed charging design selection in all three communities, but the form of the change depended on local operating patterns. In Khayelitsha, the agent-based model selected substantially more installed power; in Bellville, it selected a different connector allocation at the same total installed power; and in Wynberg, it selected both a different allocation and more installed power. Across all three communities, the replay-selected designs performed worse in terms of EV passenger throughput when evaluated in the agent-based model. These results suggest that replay-based analysis remains useful for evaluating electrification feasibility under observed mobility, but that it may not identify the charging design that best supports passenger service when drivers can adapt. Agent-based simulation can represent these additional operational effects, provided its demand assumptions, behavioural calibration and movement representation are assessed explicitly.

References

[1] NASAC, IAP, Decarbonisation of Transport in Africa: Opportunities, Challenges and Policy Options, Technical Report, Network of African Science Academies (NASAC) and InterAcademy Partnership (IAP), Nairobi, 2024.

[2] S. O. Bankole, D. L. Donaldson, D. Dissanayake, E. J. Sakamoto Ferranti, Electrifying paratransit in sub-Saharan Africa: Perspectives from practitioners, Transportation Research Part D: Transport and Environment 150 (2026) 105113. doi:10.1016/j.trd.2025.105113.

[3] M. J. Booysen, C. J. Abraham, B. G. Pretorius, A system-of-systems framework for planning the electrification of paratransit, Energy Strategy Reviews 62 (2025) 101892. doi:10.1016/j.esr.2025.101892.

[4] Sebenza, Minibus taxi wi-fi GPS telemetry dataset, 2025.

[5] J. T. Sello, M. J. Booysen, Daily Distance, Activity, Mobility Shifts among Observed Minibus Taxis in Gauteng, 2026. doi:10.2139/ssrn.7505038. arXiv:7505038.

[6] City of Cape Town, Comprehensive Integrated Transport Plan 2023–2028, Council-approved plan, City of Cape Town, Transport Planning and Network Management Department, Cape Town, 2023. URL: https://resource.capetown.gov.za/documentcentre/Documents/City%20strategies,%20plans%20and%20frameworks/CITP_2023-2028.pdf, council resolution C 10/05/2023.

[7] City of Cape Town, Taxi routes in Cape Town, Dataset, Western Cape Government Spatial Data Warehouse, 2021. URL: https://gis.westerncape.gov.za/portal/home/item.html?id=6c85897a736649018f230a2c5f8032c9, originally published on the City of Cape Town Open Data Portal, 18 May 2021. Accessed 15 October 2026.

[8] B. Pretorius, J. Strauss, M. Booysen, Grid and mobility interdependence in the eventual electrification of operational minibus taxis in cities in sub-Saharan Africa, Energy for Sustainable Development 79 (2024) 101411. doi:10.1016/j.esd.2024.101411.

[9] Higer Bus Company Limited, Higer h5c ev, Product brochure and technical specification sheet, 2022.

[10] J. T. Sello, C. J. Abraham, S. Lacock, M. J. T. Booysen, Empirical performance assessment of South Africa's first commercial electric minibus taxi, African Transport Studies 4 (2026) 100117. doi:10.1016/j.aftran.2026.100117.

[11] R. Behrens, D. McCormick, D. Mfinanga (Eds.), Paratransit in African Cities: Operations, Regulation and Reform, 1 ed., Routledge, London, 2015. doi:10.4324/9781315849515.

[12] M. J. Booysen, B. G. Pretorius, J. T. Sello, A sustainable transition to electric minibus taxis in Cape Town's paratransit, Nature Sustainability (2026). doi:10.1038/s41893-026-01808-9.

[13] J. Sello, M. Booysen, Agent-based simulation of electric minibus taxis: Framework and application to sub-Saharan Africa's paratransit, Energy for Sustainable Development 93 (2026) 102011. doi:10.1016/j.esd.2026.102011.

[14] Western Cape Provincial Parliament, Transport and public works, https://www.wcpp.gov.za/?q=node/11846, 2022-04-08, 2022-04.

[15] J. C. Thiele, W. Kurth, V. Grimm, Facilitating Parameter Estimation and Sensitivity Analysis of Agent-Based Models: A Cookbook Using NetLogo and R, Journal of Artificial Societies and Social Simulation 17 (2014) 11.

[16] S. Watanabe, Tree-Structured Parzen Estimator: Understanding Its Algorithm Components and Their Roles for Better Empirical Performance, 2026. doi:10.48550/arXiv.2304.11127. arXiv:2304.11127.

Appendix A. Validation of the vehicle communities

Appendix A.1. Similarity graph and partition

Vehicle communities are detected with a multi-view similarity graph. Each vehicle is described by four views of non-negative feature weights, which are computed from the stop events and edges inside the study area on valid weekdays: the dwell hours at each non-home node, the vehicle-days on each network edge, the dwell hours at each formal node, and the vehicle-days with each pair of consecutive anchor dwells (origin-destination pairs). The features of each view are weighted with term frequency-inverse document frequency (TF-IDF) [**?**]:

$$\tilde{x}_{vf}^{(m)} = \ln\left(1 + x_{vf}^{(m)}\right)\left(\ln\frac{1+n}{1+\mathrm{df}_f} + 1\right), \tag{A.1}$$

where $x_{vf}^{(m)}$ is the weight of feature f for vehicle v in view m, n is the number of vehicles, and df_f is the number of vehicles that use feature f. Each weight vector is subsequently normalised to unit length, such that a hub used by the whole fleet contributes less to the similarity of two vehicles than a stop shared by only a few. The combined similarity $S_{uv} \in [0, 1]$ of vehicles u and v is the mean cosine similarity over the views that both vehicles have. A similarity graph joins each vehicle to its 15 most similar vehicles, and the Leiden algorithm [**?**] partitions the graph by maximising the modularity with a resolution parameter γ [**? ?**]:

$$Q_\gamma = \frac{1}{2W}\sum_{u,v}\left[A_{uv} - \gamma\frac{k_u k_v}{2W}\right]\mathbf{1}\left[c_u = c_v\right], \tag{A.2}$$

where A_{uv} is the edge weight between vehicles u and v, k_u is the strength of vehicle u, W is the total edge weight, c_u is the community of u, and $\mathbf{1}[\cdot]$ is the indicator function. A larger γ produces smaller communities.

For each resolution $\gamma \in \{0.25, 0.5, 0.75, 1, 1.25, 1.5, 2, 2.5, 3\}$, the Leiden algorithm is run with 50 seeds, and the seed partitions are combined into a consensus partition [**?**]. A resolution is eligible when its median seed partition contains at least two communities of five or more vehicles, and when its consensus partition retains at least 90% of corridor use within one community, where a corridor is an undirected pair of consecutive anchor dwells that recurs on at least three vehicle-days and two dates. The latter condition encodes the operational meaning of a community, namely vehicles that share corridors in a common operating area. Of the eligible resolutions, the most stable resolution, as measured by the mean pairwise adjusted Rand index (ARI) [**?**] of its seed partitions, is selected. Vehicles in communities of fewer than five vehicles are not assigned to a community. The existence of the communities is tested against a null model of randomised fleets [**?**], and their seed stability, temporal stability and agreement across views are assessed in this appendix.

Appendix B. Network construction and coverage

Appendix B.1. Stop events and homes

Stop events are extracted from the GPS trajectories with an anchored-dwell algorithm that tolerates isolated positioning errors. A candidate stop is anchored at the median position of its first three inlier points, and a later point is an inlier when it lies within 30 m of the anchor and the pairwise diameter of the inliers remains at or below 60 m. One isolated outlier is permitted when the subsequent point returns to the candidate, whereas a second consecutive outlier, or a reporting gap of more than 300 s, ends the candidate. A candidate becomes a stop event when it contains at least three inlier points, has an outlier share of at most 10%, and lasts at least $T_{\min} = 180$ s, a value that was selected from a sweep of 30 to 600 s. At the sampling interval of about 80 s, three inlier points span at least about 160 s. Consequently, stops shorter than about 2 to 3 minutes, which include most of the boarding and alighting stops along a route, cannot be observed in the data. The location of a stop event is the median of its inlier positions.

The home base of each vehicle is identified with a boundary-point rule. The first and last points of each valid vehicle-day are collected, the boundary point with the most other boundary points within $R_{\mathrm{home}} = 300$ m is selected as the candidate, and the home location is the median of the boundary points within R_{home} of the candidate. A vehicle has a home when both the first and the last points of at least 75% of its valid vehicle-days lie within R_{home} of this location; otherwise, its most frequent first point is used as its start location. The sensitivity of the rule was assessed for radii from 150 to 1,000 m and minimum shares of 0.6, 0.75 and 0.9.

Appendix B.2. Nodes, edges and review

A single operational network is constructed for the whole study area and is used by all communities. Rather than representing the full road network, the network represents the locations at which taxis stop in an operationally important way, the fixed infrastructure that the fleet uses, and the links between these locations, such that nodes are derived from evidence of stopping and edges from evidence of driving. The eligible stop events (inside the study area, on valid weekday vehicle-days, and not at a home unless that home is at infrastructure) are counted in two square grids of 100 m cells with offsets of 0 and 50 m, so that a rank on a cell boundary is not split into two weakly supported cells. A cell is selected when it contains stop events from at least $N_{vd} = 5$ vehicle-days and $N_v = 3$ distinct vehicles. This support rule replaces the classifier of an earlier stop-network method and achieved a higher precision and F1 score than the classifier on held-out labelled cells from Gauteng. The selected cells are merged into support nodes by connected complete-linkage clustering with a maximum diameter of $D_{\max} = 200$ m, which prevents long spatial chains, and each stop event is associated with one node only.

Formal nodes are obtained from the City of Cape Town minibus-taxi route termini, the taxi ranks and railway stations in OpenStreetMap (OSM), and one large rank that these sources omit, namely the Site C rank in Khayelitsha, which was confirmed on satellite imagery (Section 3.3). Formal nodes within 30 m of each other are merged into one node, a formal node is retained only when the fleet stops there, and a support node within the snap distance $S = 100$ m of a formal node joins it and adopts its location. Home bases that are not at infrastructure become home nodes, which vehicles can reach but cannot traverse.

Since a moving vehicle covers a median distance of 435 m between consecutive GPS points, a vehicle can pass through several nodes between two stop events without a detected stop. A pass is therefore recorded when the straight line between two consecutive GPS points comes within $r_{\text{pass}} = 100$ m of a node, with the pass time and cumulative distance interpolated at the point of closest approach. Passes split the movement between two stop events into hops, but they do not count towards the support of a node, so that nodes remain defined by where taxis stop. The network contains three kinds of directed edges: observed edges, which join the two nodes of a hop and are retained when at least two vehicle-days use them; derived reverse edges for observed edges without an observed counterpart; and close-node links in both directions between nodes within $D_c = 300$ m of each other, which allow short movements between nearby nodes without long detours. All edge lengths are shortest road paths on the drivable OSM network, such that one-way streets are respected.

Rather than relying solely on automatic rules, the network is subjected to a single manual review of 211 items, which comprise candidate cells with high vehicle-day support but few vehicles, edges with implausible ratios of road length to straight-line length or implausible speeds, and conflicts between support nodes and formal nodes. Each item was reviewed in an interactive map that plays back the relevant vehicle-days, and 2 of 30 candidate cells, 153 of 168 flagged edges and all 13 conflicts were accepted. The resulting network comprises 726 nodes, of which 18 are ranks, and 98.9% of the nodes lie in the largest strongly connected component. The nodes capture 76% of the eligible stop events and 72% of their dwell time, and 66.5% of the distance driven between stop events runs on network edges.

Appendix B.3. Route library and route sets

Since the intermediate nodes of a taxi vary from trip to trip between the same terminals, the observed trips are compressed into representative routes. For each ordered terminal pair, the distinct node sequences are processed in order of decreasing support, and a sequence joins the most similar existing variant when the Jaccard similarity of their node sets [**?**] is at least $\theta = 0.5$; otherwise, it founds a new variant. A variant qualifies as a route when its trips occur on at least three vehicle-days and two dates. The representative of a qualifying variant is its medoid, i.e., the observed trip with the highest mean similarity to the other trips of the variant, and the stops of the route are the terminals and stop nodes of the medoid, together with each pass that at least half of the variant's trips visit. This majority rule removes the noise of parallel roads while retaining the nodes at which taxis consistently search for passengers. All route nodes are possible stops, and whether a driver stops at a node is a behavioural decision in M_{A} (Section 4.4).

The route library is built from the trips of the whole fleet, because a corridor is a property of the operation and more vehicles provide more support. Each route receives a reverse counterpart, which is either the most similar qualified route in the opposite direction or a derived route with the reversed node sequence. Additionally, each loop is split at its turning node (the node farthest from the origin) into an outward and

a return half, so that a taxi can turn at the turning node and start a new run there. The resulting library comprises 586 qualified routes and 503 derived reverse routes.

The eligible routes of a driver are the smallest set of its routes, ordered by their share of its trip distance on qualified routes, that covers 80% of that distance, together with the reverse counterparts of these routes and the two halves of each eligible loop. In the selected communities, routes shorter than 5 km in the network are excluded, since a route whose node sequence retains only a few nodes can be much shorter in the network than its observed trips and would cause an agent to drive too little. A vehicle with at least one eligible route becomes an agent of both models. The routes carry no time of day, and the direction and timing of demand are instead handled by the demand model (Section 4.5). Finally, a route is treated as a single unit of operation in M_{A}: once a driver has started a route, it cannot charge at a rank along that route. Consequently, an EV does not select a route that it cannot complete from a full battery (Section 4.4).

Appendix C. Agent-based model specification

Appendix C.1. Population and shifts

The drivers of a community comprise its agents and their synthetic ICE copies. Each agent keeps its home node, its eligible routes and its idle locations, which are the terminals of its routes. Since the observed vehicles are only a sample of the taxis that operate from a hub, each agent receives $\lfloor k-1 \rfloor$ or $\lceil k-1 \rceil$ ICE copies from a seeded Bernoulli draw, such that the population is on average k times the number of agents, where k is the scale factor of Section 4.5. A copy adopts the routes, idle locations and shift pattern of its agent, its home is drawn from the five network nodes nearest to the agent's home, and the start and end of its shift are moved by a common normal draw with a standard deviation of 30 min. This construction rests on the assumption that the observed agents are representative of the taxis that load at the hub of the community, both in the routes that they drive and in their working hours. The resulting populations comprise 1,358, 1,673 and 686 drivers in Khayelitsha, Bellville and Wynberg, respectively, of which the electrified agents form only 2.6% to 6.6% (Table 6).

Each agent has a typical-weekday shift, of which the start and end times are normally distributed with the mean and standard deviation of the first and last movements on its calendar days in the calibration period, and each day is a working day with the observed probability of activity. Two calibrated parameters, the shift spread ψ and the shift end offset δ, scale the standard deviations and move the mean end time, respectively. The offset is necessary because the observed end of a shift is the arrival at home, whereas the simulated end is the time at which the driver starts to return home. In Khayelitsha, a last-run rule additionally prevents a driver from starting a route that it expects to end after its shift end, because too many agents otherwise remained active late at night.

Appendix C.2. Travel time

The travel time of an edge ℓ that a driver enters in time window w is modelled as follows:

$$T_\ell = \frac{s_\ell}{80 \text{ km/h}} + X_{\ell w}, \qquad X_{\ell w} = \begin{cases} 0, & \text{with probability } \pi_{\ell w}, \\ c_w Y, \quad \ln Y \sim \mathcal{N}\left(\mu_{\ell w}, \sigma^2_{\ell w}\right), & \text{otherwise}, \end{cases} \tag{C.1}$$

where s_ℓ is the road length of the edge, $\pi_{\ell w}$ is the share of observed hops without excess time, $\mu_{\ell w}$ and $\sigma_{\ell w}$ are the parameters of a log-normal fit of the observed excess times [**?**], and c_w is an optional excess factor. The four time windows are the night (19:00 to 06:00), the AM peak (06:00 to 09:00), the midday (09:00 to 15:00) and the PM peak (15:00 to 19:00). The fit uses the hops of the whole fleet on the calibration days, excluding hops slower than 5 km/h, since such hops contain a stand rather than driving and would make the simulated vehicles much slower than the observed vehicles. Edges with fewer than five retained hops receive a fixed excess from the pooled speed of the window, reverse edges copy the model of their observed counterpart, and close-node links use the median speed of the short observed edges. In Khayelitsha, where the simulated agents drove faster than the observed vehicles in every window, the excess is scaled by a factor c_w for each window to match the observed median speed. Drivers travel along shortest paths by road length [**?**], and the model does not represent congestion. Only the 18 formal ranks of the network act as engine ranks with loading queues.

Appendix C.3. Route choice

A driver that is not on a route first returns home when its shift has ended. Otherwise, an EV driver consults its charging policy, after which the driver chooses between its routes and idling with a multinomial logit model [**?** **?**]:

$$\Pr(j) = \frac{\exp\left(U_j/\tau(t)\right)}{\sum_i \exp\left(U_i/\tau(t)\right)}, \tag{C.2}$$

where U_j is the utility of route j or of idling, and $\tau(t)$ is a temperature that decreases from 1 to 0.1 during an initial exploration period of ten days, such that drivers initially sample all of their routes [**?**]. The utility of a route is a weighted sum of normalised features: the waiting requests at the current node that the route can carry, the competing drivers at the node on the same route, and the remembered distance, passengers and passenger distance per run of the route in the current interval of the day. The utility of idling comprises a bias β_0, a demand term with weight β_h that increases as the driver's remembered demand in the current interval decreases, and a small income term, such that drivers idle more during intervals that they remember as quiet. The route weights are fixed at the values of an earlier calibration of the model, whereas β_0 and β_h are calibrated (Section 4.6).

Appendix C.4. Boarding and memory

A passenger boards only a taxi that is at the origin node, has a free seat and serves a route on which the destination follows the origin; at a rank, the first compatible taxi in the loading queue is selected. Otherwise, the request waits and expires after 2 h. Passengers do not walk to other nodes or switch to other modes. The memory of each driver records the distance, duration, passengers and passenger distance of each completed route run, as well as the queue length, waiting time and charging time of each charging visit, as streaming statistics in 30-minute intervals of the day. A driver therefore learns where and when its routes carry passengers.

Appendix C.5. Run horizon

A cold run starts with empty memories and runs a warm-up period followed by 14 evaluation days [**?** **?**]. The warm-up length of each community, between 49 and 91 days, is the first day after which the seven-day means of the served passengers, their share of the requests, the occupancy and the route-choice shares remain within a tolerance of their long-run means on three seeds, rounded up to whole weeks. To reduce the computational cost, the warm-up is run once, and the resulting driver memories are saved as a warm start. Each subsequent run loads these memories, disables exploration, and runs 14 settling days followed by 14 evaluation days.

Appendix C.6. Charging and reserve rules

To prevent avoidable stranding, an EV maintains a reserve of $0.2\,C$. It does not select a route that a full battery cannot complete together with the drive from the last stop to the nearest station and the reserve. Moreover, it starts a route only when its energy covers the drive to the first stop, the route, the drive to the nearest station and the reserve, and otherwise charges first, and it charges before returning home when it would otherwise arrive below the reserve. At a station, a driver takes a connector only while it still needs to charge: before its shift end, while it is below its charge target or the energy needed for its cheapest route, or while its policy requests charging; and after its shift end, only while it would otherwise arrive home below the reserve. Home charging at 3.7 kW is applied for the time at home. An EV that nevertheless strands loses the passengers on board and is restored at home with a full battery at the next day start. These rules are corrections of the engine that replace its earlier charging behaviour without a setting to revert to it. Since ICE drivers never charge, the corrections do not change the ICE runs, which was verified by reproducing the logs of a frozen ICE run exactly.

Appendix C.7. Passenger demand

A departure is an anchor-to-anchor trip of an agent on the calibration days that is at least 1 km long and leaves the catchment of its origin. The catchment is defined as all nodes within 300 m of the same facility of the Comprehensive Integrated Transport Plan (CITP) of the City of Cape Town or, if no such facility exists, the origin node alone, which removes short movements within a rank complex. Since the stop rates, slow-moving shares and speeds of the observed trips do not distinguish loaded from empty legs at this sampling interval, the direction of demand is taken from survey data. Each departure receives a load factor λ, which equals 1 except for departures in the counter-peak direction of a corridor during the AM peak (06:00 to 09:00) and in the AM-peak direction during the PM peak (15:00 to 19:00). For these departures, λ equals the ratio of counter-peak to peak vehicle trips on the corridor in CITP Table 3-18, which ranges between 0.09 and 0.39, or the mean ratio of 0.21 for corridors without a survey. The peak direction of a corridor is the surveyed direction with more AM-peak trips or, for corridors without a survey, the direction from its residential end, which is identified from OSM land use. Since the agents drive only their eligible routes in M_{A}, the departures of each agent are moreover redistributed onto its eligible routes in proportion to their load factors with a weight ω_d, such that each agent retains its total load and the supply and demand remain consistent with what the agents drive.

The scale factor is the ratio of the full population of taxis that load at the main hub to the agents:

$$k = \frac{B}{Q\,L}, \qquad L = f \sum_{d \in \mathcal{H}} \lambda_d\, \omega_d, \qquad f = \frac{n_{\mathrm{active}}}{n_{\mathrm{valid}}\,|\mathcal{D}_c|}, \tag{C.3}$$

where B is the daily number of boardings at the CITP facility of the main hub (Section 3.3), $\mathcal{H}$ is the set of agent departures from the hub catchment on valid vehicle-days, and L is the number of full loads of the agents per day. The day factor f converts counts on valid vehicle-days to counts per calibration day, where n_{active} and n_{valid} are the active and valid agent vehicle-days and $|\mathcal{D}_c| = 6$ is the number of calibration days, which corrects for the strong variation of the valid share between dates. This formulation assumes that the routes of the community represent all routes of its hub and that departures in the peak direction and off-peak depart full. The resulting scale factors are given in Table 6. As a plausibility check, k is compared with an upper bound that is derived from the operating licences in the provincial route register. The Bellville value exceeds this bound, which is discussed as a limitation in Section 6.3.

Each departure with load factor λ on a route with node sequence $s_0, s_1, \ldots, s_m$ generates $Q\lambda$ terminal boardings at s_0, which alight at the downstream nodes in proportion to the node weights $w_n = 1 + c_n$, where c_n is the number of agent stop events at node n on the calibration days. The floor of one gives every route node some alightings, since short stops cannot be observed in the data. A further $Q\lambda(1-\alpha)/\alpha$ passengers board and alight between the en-route nodes in proportion to $w_{s_i} w_{s_j}$, such that a share $\alpha = 0.8$ of all passengers boards at the terminal. The terminal share α cannot be identified from the data and is therefore a judgement value. The daily number of requests of each origin-destination pair is given by:

$$R_{od} = k\, f \sum_{d} \omega_d\, r_{od}(d), \tag{C.4}$$

where the sum covers the eligible departures d on valid vehicle-days and $r_{od}(d)$ is the number of requests that departure d generates on the pair. Since k is derived from the same departures, load factors and weights, the terminal requests at the hub equal the facility boardings by construction. Moreover, demand is generated only for servable pairs, i.e., pairs that lie in order on an eligible route of an agent.

The daily requests of each pair are distributed over the hours of the day according to the profile of its direction class (peak, counter-peak or no direction). In Wynberg, the profile is the load of the agents' eligible departures in each hour, including the evening trips home, without which the profile would have almost no evening peak. In Khayelitsha and Bellville, the departures of the agents are almost uniform over the day, whereas their moving time shows clear morning and evening peaks, i.e., at the peaks, the taxis move longer rather than more often. The profile of these communities is therefore the observed moving time of the agents in each hour, shifted earlier by half the median trip duration to approximate the boarding time and weighted by the mean load factor of the class in the period. Wynberg retains the departure-based timing with which it was calibrated and accepted. In the engine, the number of requests of each pair in each 15-minute interval is drawn from a Poisson distribution with uniformly distributed request times, which yields a non-homogeneous

Poisson process with a piecewise-constant rate [?]. The same profile applies to every simulated day. The resulting demand levels are given in Table 6, and the demand is audited against the corridor survey counts and the published boardings at facilities outside the communities.

Appendix D. Calibration targets and search

Appendix D.1. Targets and fit

The calibration targets are computed from the observed agents on the calibration days and comprise four families: the share of agents moving in each hour of the day; the 10th, 50th and 90th percentiles of the daily distance; the same percentiles of the daily hours moving (GPS steps at 5 km/h or faster, the same threshold that separates a stand from a hop in the travel times); and the same percentiles of the coefficient of variation of each agent's daily distance. Additionally, at least 95% of the requests must be served (a constraint), and the median seat occupancy should be as high as possible up to 0.6 (a goal), which reflects full departures from the rank and partly empty returns. The noise σ_i of each target is the standard deviation of its statistic over 500 bootstrap resamples of whole calibration dates [?], which preserves the dependence between vehicles on the same date. The fit of target i is given by:

$$z_i = \frac{\left| y_i^{\mathrm{sim}} - y_i^{\mathrm{obs}} \right|}{\max\left(\sigma_i,\ \epsilon_i\right)}, \tag{D.1}$$

where y_i^{sim} and y_i^{obs} are the simulated and observed values of the target, and ϵ_i is a floor (0.01 for the movement share, 5 km, 0.02 for the coefficient of variation and 0.25 h) that prevents targets with very small noise from dominating the fit. The loss of a family is the mean fit of its targets, and the loss of a run is the weighted mean of the family losses.

Appendix D.2. Search

Following [15], the calibration proceeds in two stages. First, hand tuning from the warm start locates a starting point within the region of plausible operation and identifies gaps that no behavioural parameter can close. Such gaps required changes to model rules, namely the travel-time scaling and the last-run rule in Khayelitsha. Second, a search strategy is frozen in a versioned file whose SHA-256 digest is recorded. The strategy specifies nine parameters and their bounds (the idle bias and idle demand weight; the stop bias and maximum stop duration; the first-stop bias, demand weight and maximum duration; and the shift spread and shift end offset), the objectives, the constraint and the budget. The Khayelitsha strategy weights the movement family by 0.25 relative to the duty-cycle families, since a search with equal weights traded the duty cycles for the hourly shape of the day, whereas the Bellville and Wynberg strategies use equal weights and apply this priority at the acceptance. The search uses the multivariate tree-structured Parzen estimator (TPE) of Optuna [? ? ?] with two objectives, namely the minimisation of the loss and the maximisation of the occupancy capped at 0.6, subject to the served-share constraint. Each trial runs one seed from the warm start, and the search starts with a batch of 40 trials, of which 10 are space-filling start trials, followed by optional continuation batches of 20 trials. Selected points on the feasible front are subsequently validated with cold runs, which include the full warm-up, on three fresh seeds.

Appendix D.3. Assessment and hold-out

Each validated point is assessed with graded levels (good, acceptable or limited), which are expressed relative to the loss of the observed agents against their own targets over resampled calibration dates, so that the losses of the three communities are comparable. The acceptance prioritises realistic behaviour and the duty-cycle families, followed by the stability across seeds, the served share and the occupancy, and finally the hourly shape of the day. After the acceptance, a hold-out report compares the fit of the accepted model against the targets of the hold-out days (2 and 10 December) with its fit against the calibration targets. The resulting hold-out gap is assessed against the loss of the observed hold-out days against the calibration targets, which quantifies how much two days differ from six days in the data alone. The hold-out days test only the calibrated behaviour, since the network, the communities and the routes use all study weekdays.

Appendix E. Replication with a second fleet

Appendix F. Reproducibility